# SLA OR NAY?

## CAUSAL IMPACTS OF NEIGHBORHOOD-LEVEL RENT QUALITY REGULATION

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WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

PRELIMINARY RESULTS

SLA OR NAY?

# WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

PRELIMINARY RESULTS

SLA OR NAY?

DISCRETIONARY LICENSING AUTHORITY

*requirements for privately rented properties falling within a council's jurisdiction that extend national requirements for licensing*

DISCRETIONARY LICENSING AUTHORITY

ADDITIONAL LICENSING

*extends national HMO requirements to "smaller" households of 3 or more people w/ shared facilities*

## DISCRETIONARY LICENSING

DISCRETIONARY LICENSING AUTHORITY

*extends national HMO requirements to all properties for 5 years*

ADDITIONAL LICENSING

SELECTIVE LICENSING

## DISCRETIONARY LICENSING

DISCRETIONARY LICENSING AUTHORITY

*applies*

ADDITIONAL LICENSING

*to certain kinds of properties*

SELECTIVE LICENSING

*in specific areas & times*

## DISCRETIONARY LICENSING

# EASTERN SLA (1)

Addressing severe quality issues
2016-2021 in St. George West & Eastville
All privately-let properties
EPC, Electric, Gas safety inspections
£300 to £750 fee

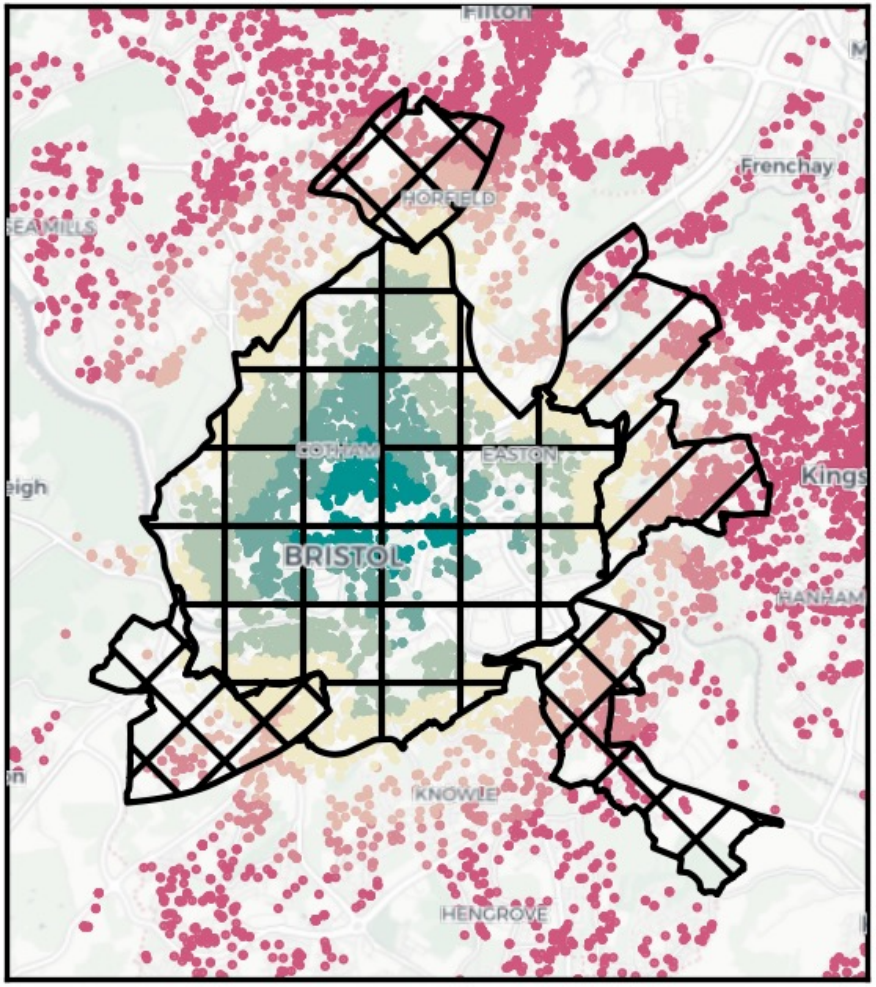
# CENTRAL ALA (2)

2019-2024 in all central wards
3+ People w/ Shared Facilities
EPC, Electric, Gas safety inspections
£900 to £1450 fee

# EXTENSION (3)

2024-2029 for different targets
EPC, Electric, Gas safety inspections
£900 to £2000 fee

## Listings near DLAs



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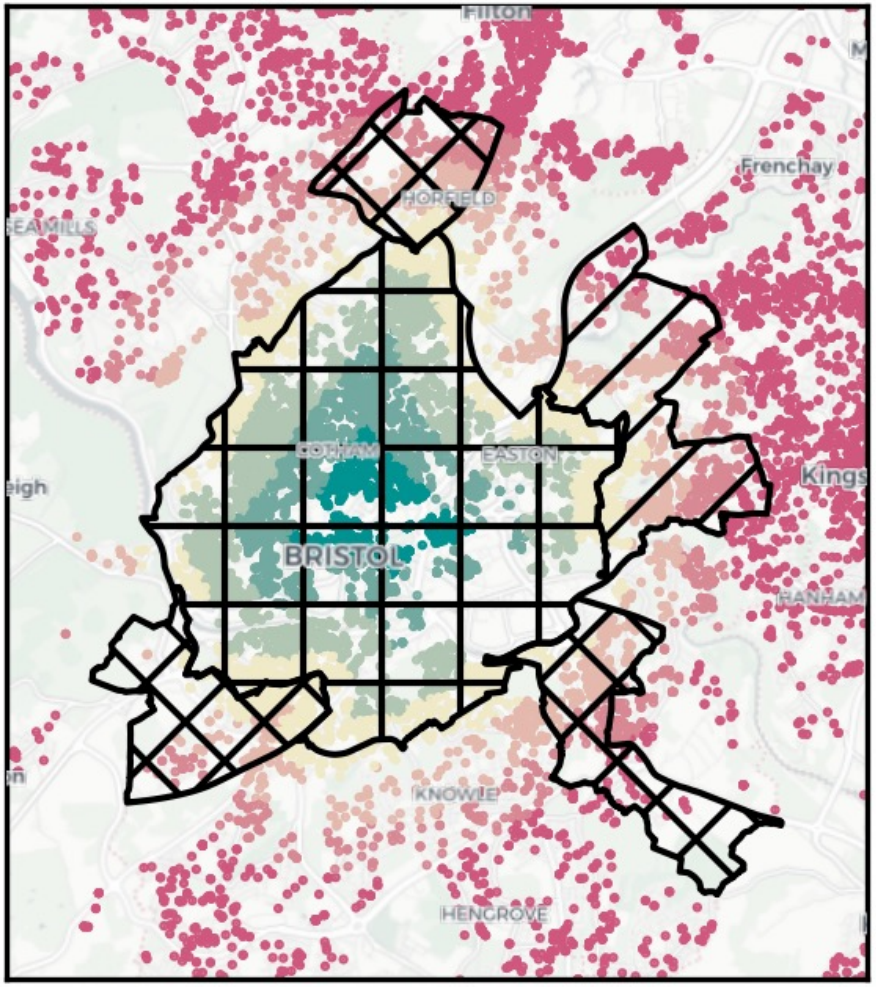
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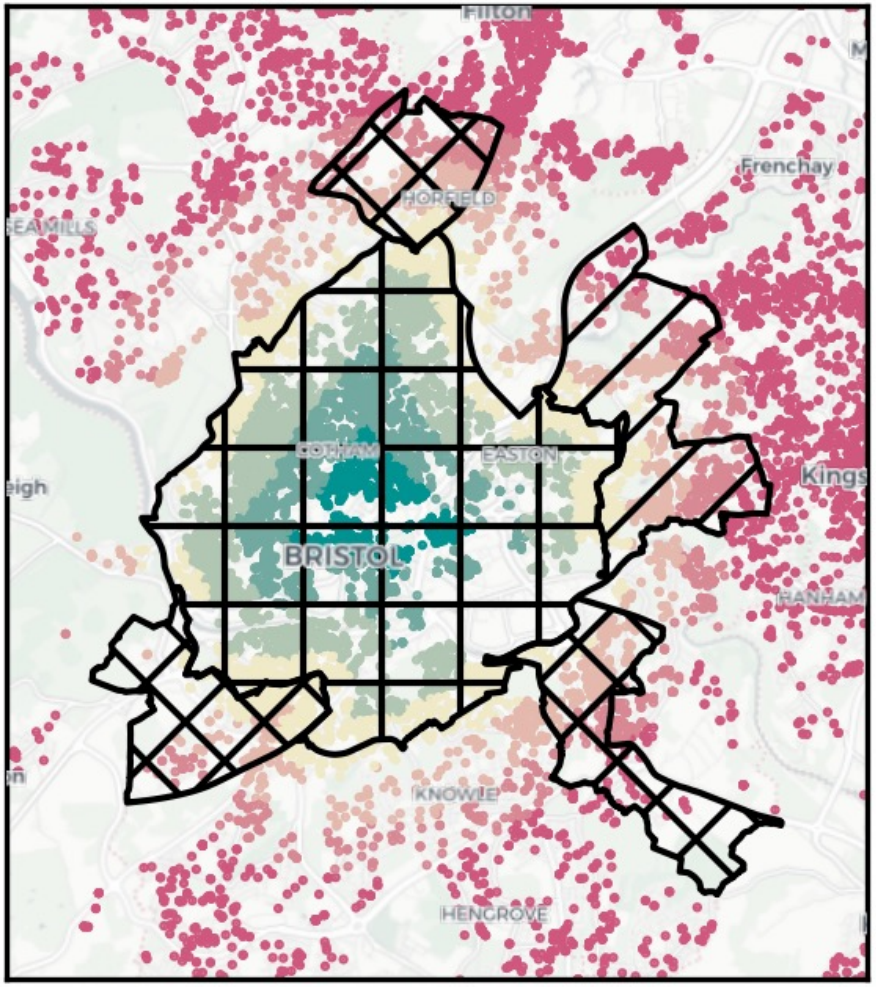
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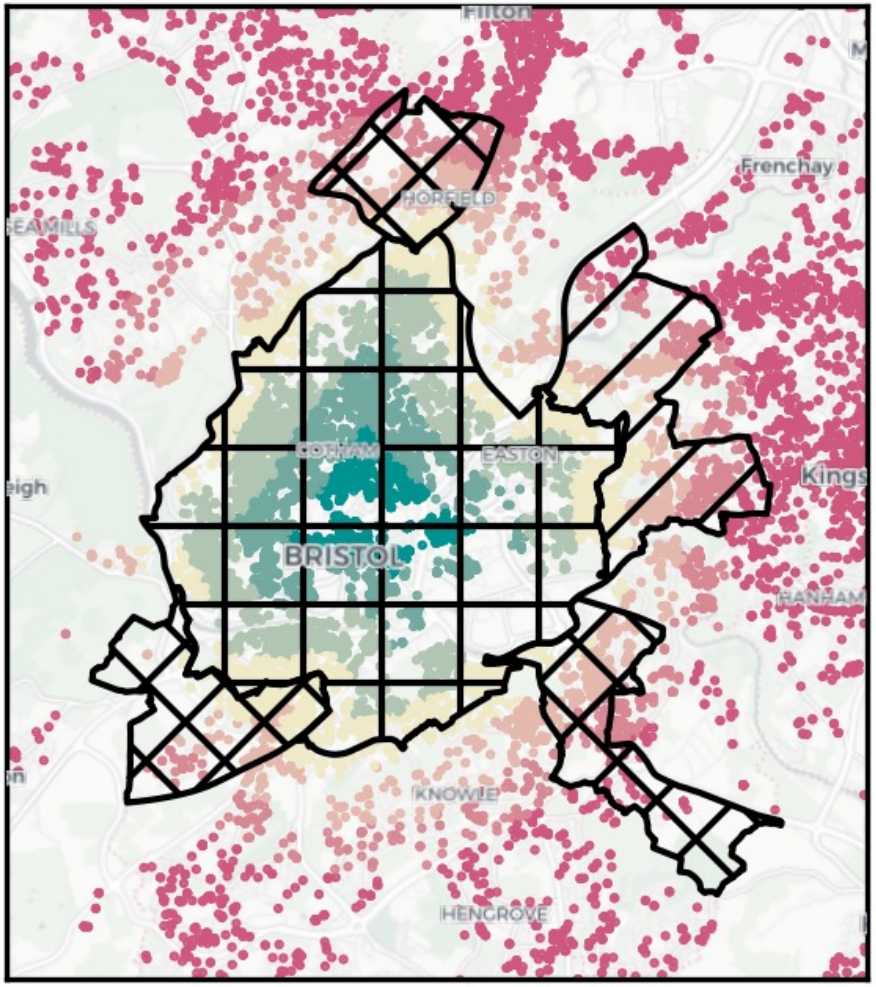
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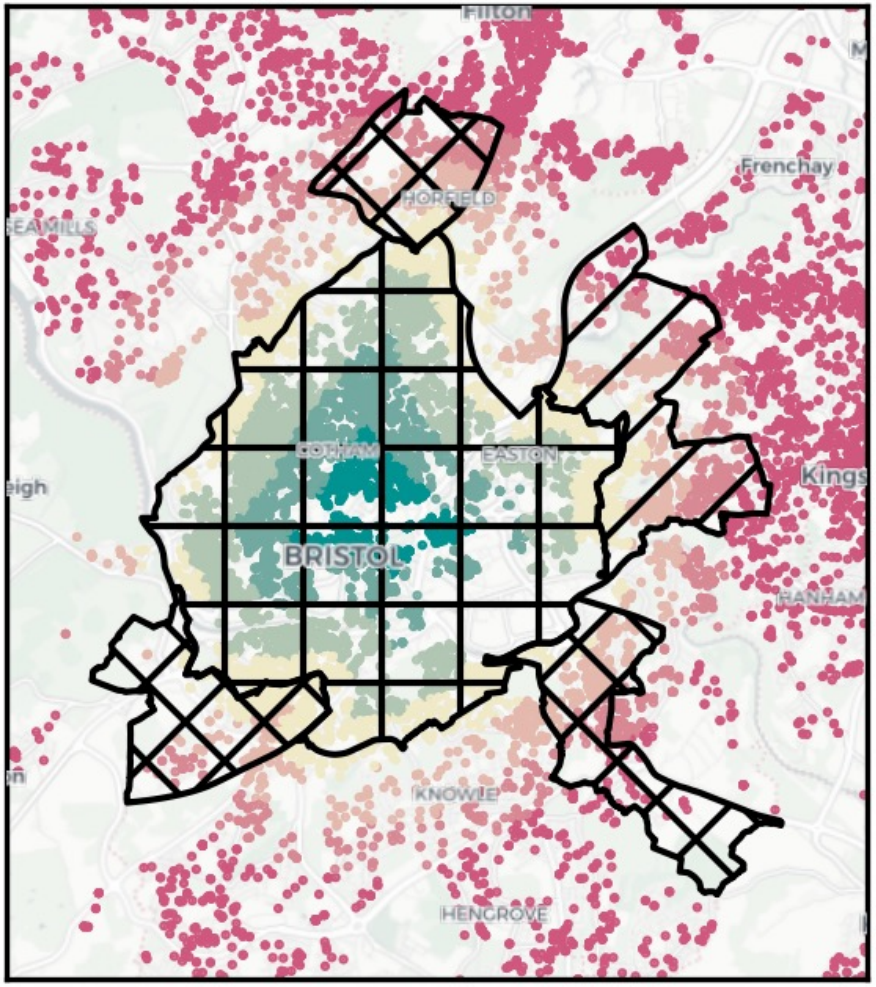
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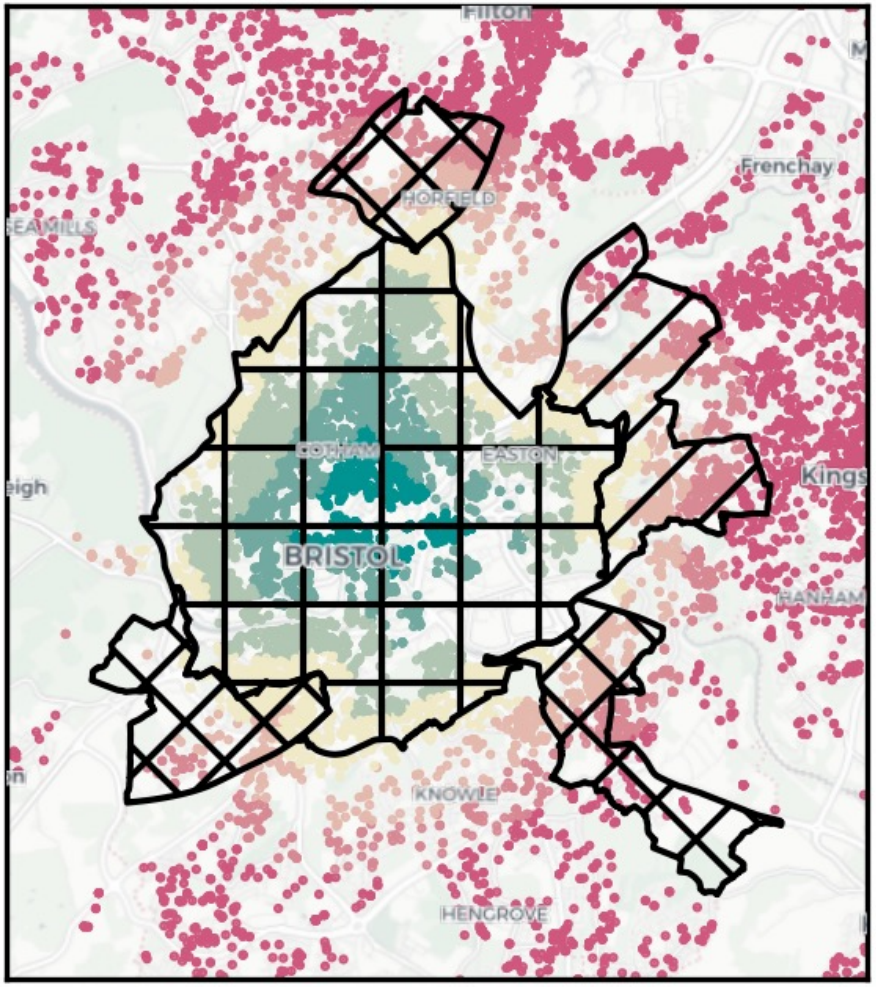
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# Listings near DLAs



City Council of Bristol (The council)
Housing Act 2004 (“The Act”): Parts 2 and 3
NOTICE OF DESIGNATION OF A CITYWIDE ADDITIONAL LICENSING
NOTICE OF DESIGNATION OF BISHOPSTON AND ASHLEY DOWN, COTHAM AND EASTON WARDS FOR SELECTIVE LICENSING



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BBC

# Bristol landlords oppose new licensing fee scheme

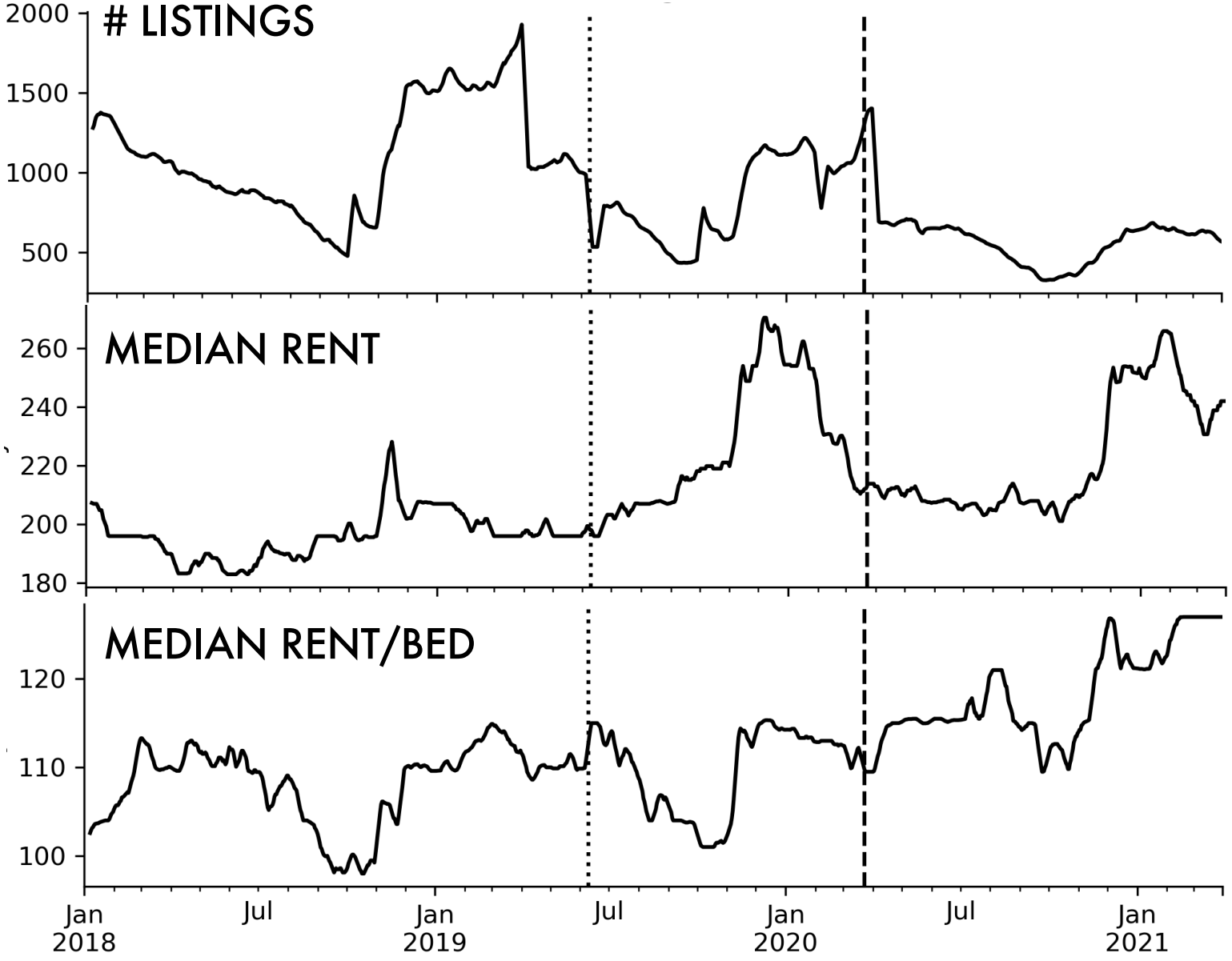
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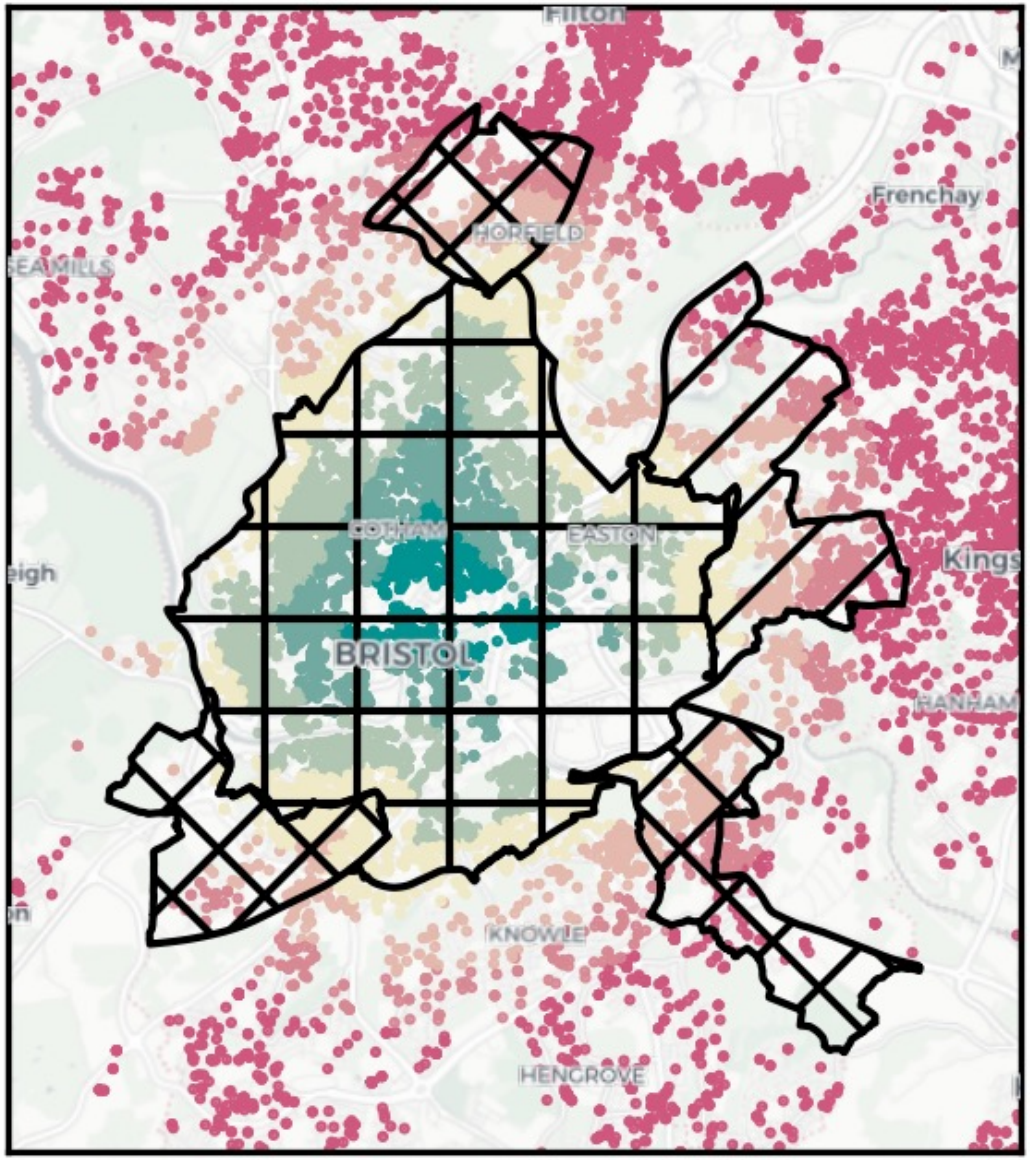
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*Fewer rentals?*

*Higher rent?*

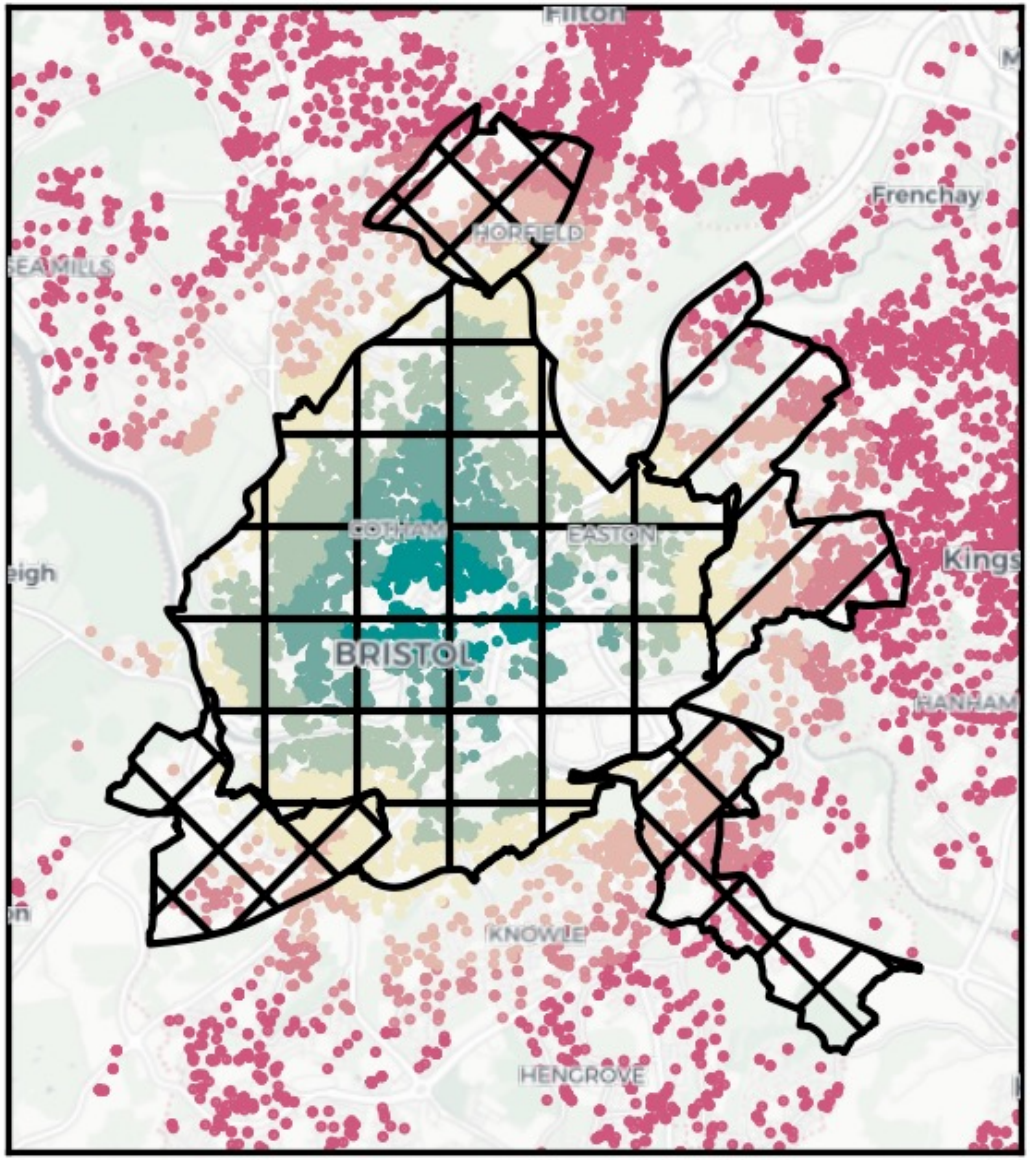
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Given the geography of these regulation areas, how can we be sure that their imposition is reflected in the “total” city rates?

Displacements **within** cities matter!

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Given the geography of these regulation areas, how can we be sure that their imposition is reflected in the "total" city rates?

Displacements **within** cities matter!

Neighborhood-level rent regulation distorts *local*, not city-wide, incentives.

We want to know: for similar properties, did licensing make a difference *on the margin?*

That is, **at the boundary of the DLA,** can we detect the difference the DLA makes?

WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

PRELIMINARY RESULTS

| | Pre-Treatment | Post Treatment |
|---|---|---|
| **Inside** | Pretreated | Treated |
| **Outside** | Pre-substitute | Substitute |

# IDENTIFICATION STRATEGIES

| | Pre-Treatment | Post Treatment |
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| **Inside** | Pretreated | Treated |
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**Treated** listings inside DLA after treatment.
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# IDENTIFICATION STRATEGY 1: SDIDc

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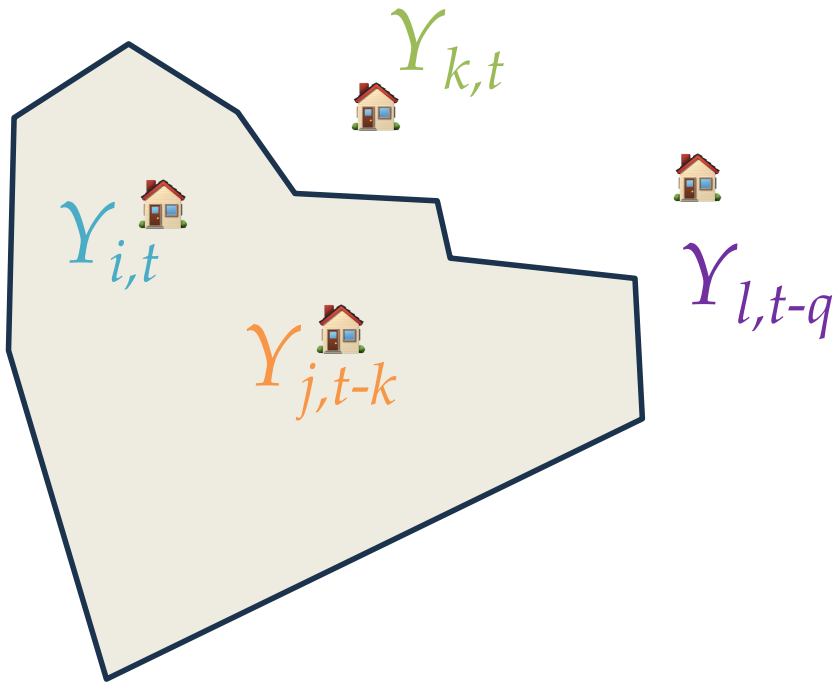
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(**T** – **S**) - (**PT** – **PS**) is the causal effect!

# IDENTIFICATION STRATEGY 1: SDIDc




**Borrow counterfactual from *just over* the boundary**

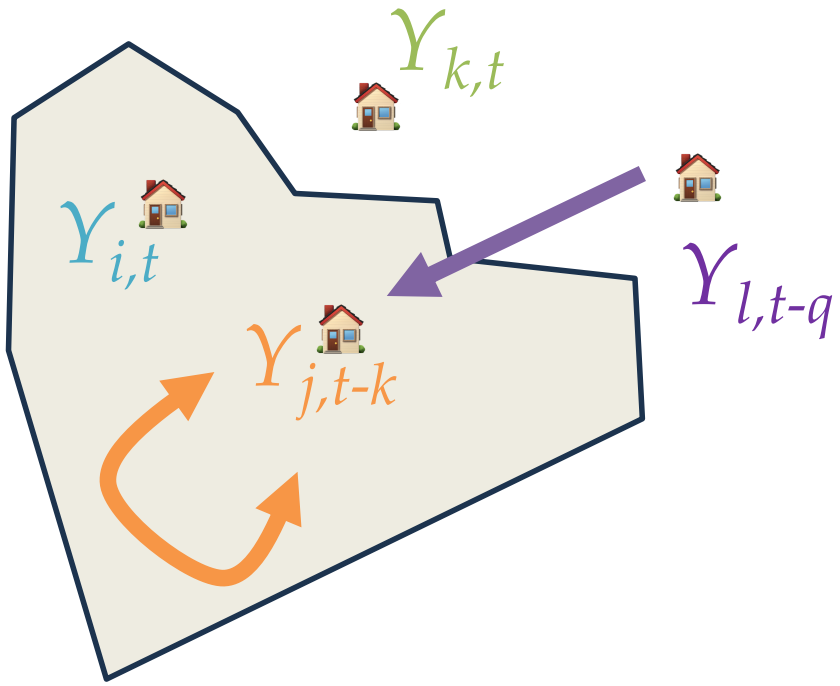
**Pretreatment model:**
fit on $Y_j$ POs from $Y_l$

**Treatment model:**
fit on $Y_i$ POs from $Y_k$

**Substitute model:**
fit on $Y_k$ POs from $Y_i$

**Pre-substitute model:**
fit on $Y_l$ POs from $Y_j$

# IDENTIFICATION STRATEGY 1: SDIDc




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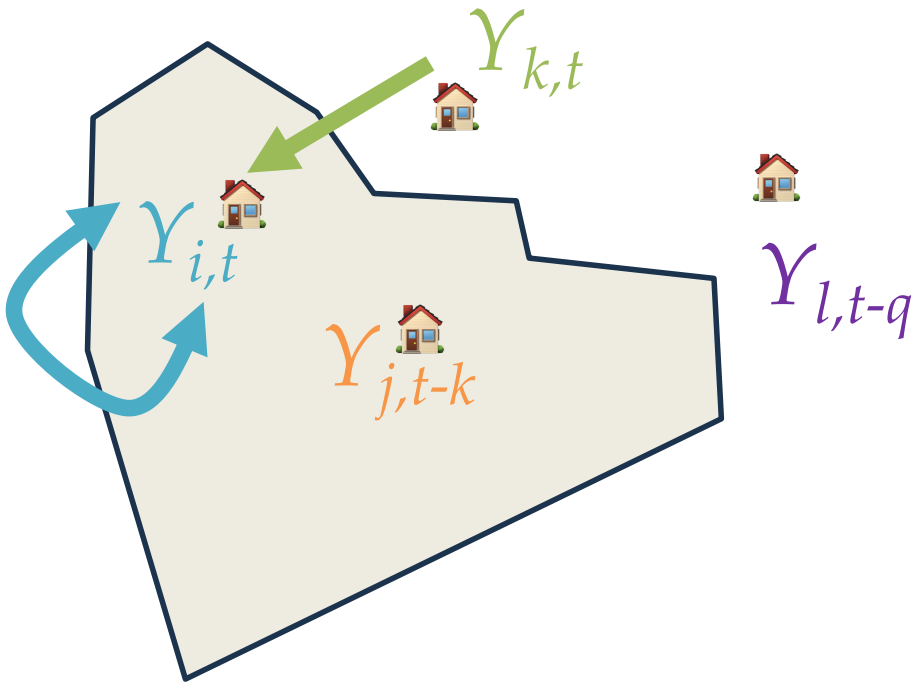
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**Discontinuity:** predicted outcome under "pre-treated" inside model vs. "never-treated" outside model

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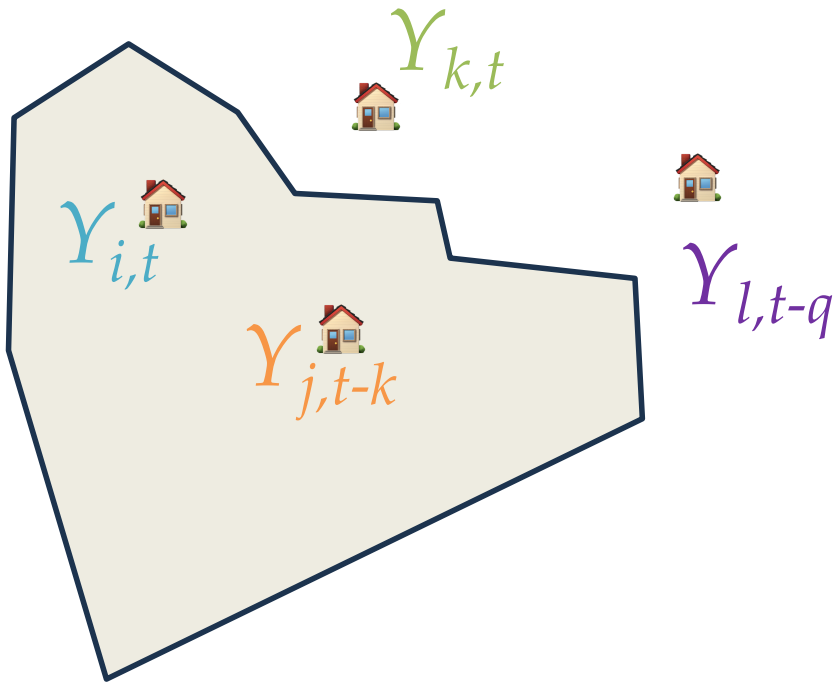
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**ATE:**
(**T** – **S**) - (**PT** – **PS**)

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(**T** – **S**) - (**PT** – **PS**) is the causal effect!

Estimated discontinuity posttreatment

Estimated discontinuity pretreatment

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**THIS IS JUST A COMPARISON ACROSS GROUP AVERAGES. WITHIN PROPERTY CHANGE IS MORE ROBUST.**

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(**T** – **S**) - (**PT** – **PS**) is the causal effect!
**T** – **S** – **PT** + **PS**
**T** – **PT** – **S** + **PS**
(**T** – **PT**) - (**S** – **PS**)

**THIS IS JUST A COMPARISON ACROSS GROUP AVERAGES. WITHIN PROPERTY CHANGE IS MORE ROBUST.**

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$$(T - S) - (PT - PS)$$
$$T - S - PT + PS$$
$$T - PT - S + PS$$
$$(T - PT) - (S - PS)$$

Estimated change inside — Estimated change outside

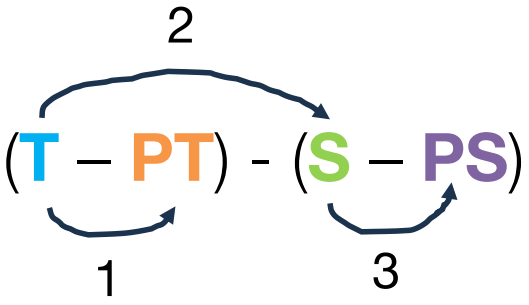
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# IDENTIFICATION STRATEGY 2: QUARTET + X

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Matches needed:

1. treated to predecessor—match back in time w/in DLA
2. treated to substitute—match contemporary to out of DLA
3. substitute to predecessor—match back in time out of DLA

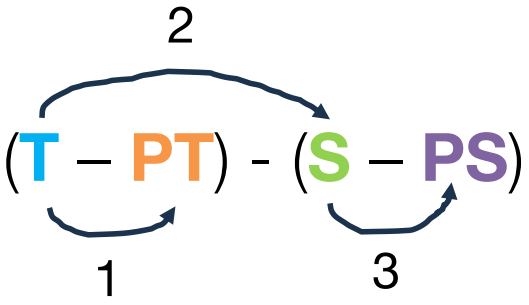


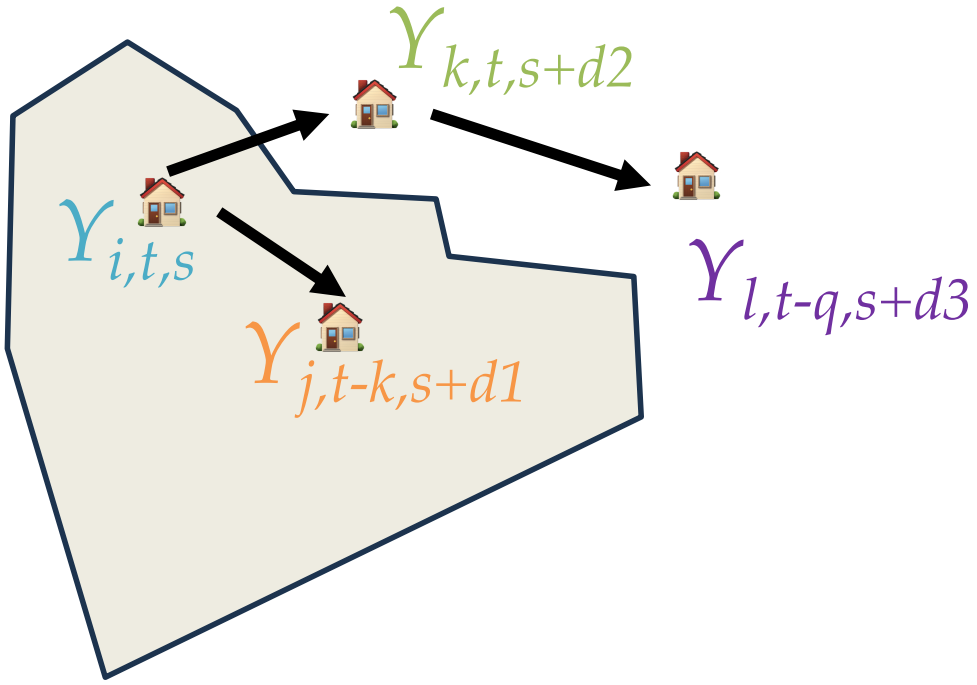

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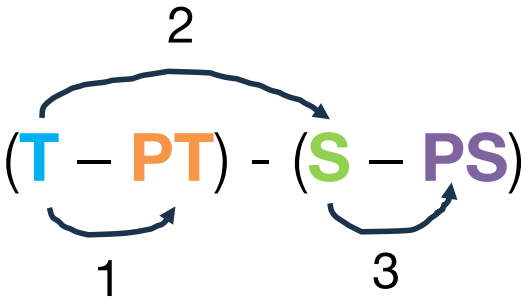



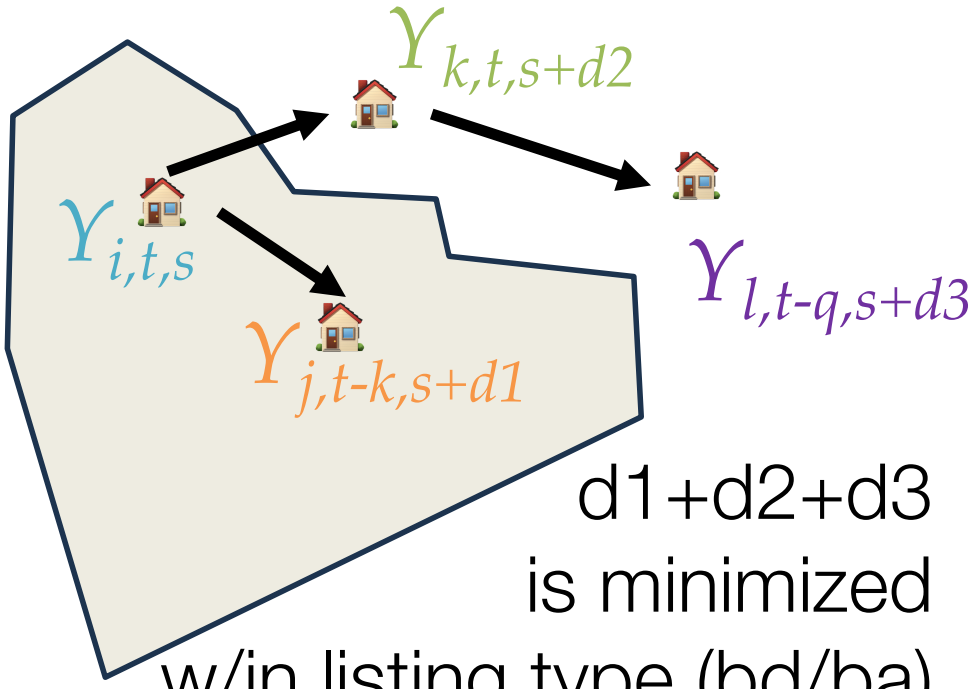

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d1+d2+d3
is minimized
w/in listing type (bd/ba)

# IDENTIFICATION STRATEGY 2: QUARTET + X

(1) T→PT (2) T→S (3) S→PS

$$\textit{minimize} \quad \Sigma\, d_{ij} t_{ij} + \Sigma\, d_{ik} s_{ik} + \Sigma\, d_{kl} p_{kl}$$

*subject to*

$$\Sigma_j\, t_{ij} = 1$$ *all treatments get one pretreatment*

$$\Sigma_i\, t_{ij} \leq 1$$ *use pretreatments at most once*

$$\Sigma_k\, s_{ik} = 1$$ *all treatments get one substitute*

$$\Sigma_l\, p_{kl} \geq \Sigma_j\, s_{jk}$$ *every selected substitute gets a presub*

$$\Sigma_i\, s_{ik} \leq 1$$ *use substitutes at most once*

$$\Sigma_k\, p_{kl} \leq 1$$ *use presubs at most once*

# IDENTIFICATION STRATEGY 2: QUARTET + X

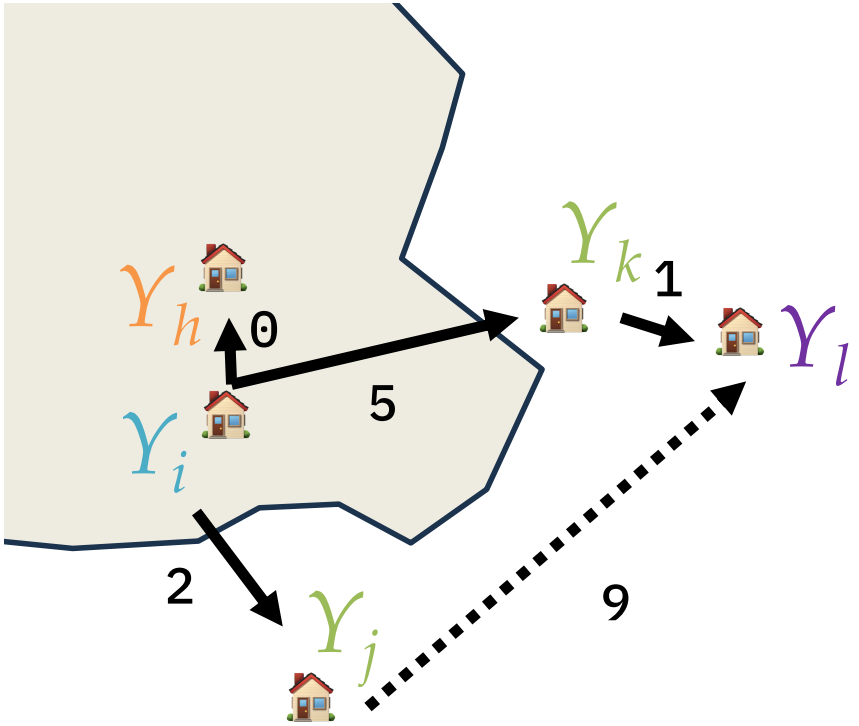
(1) T→PT (2) T→S (3) S→PS

*minimize* $\sum d_{ij}t_{ij} + \sum d_{ik}s_{ik} + \sum d_{kl}p_{kl}$

*subject to*

| | | |
|---|---|---|
| Inside subproblem | $\sum_j t_{ij} = 1$ | *all treatments get one pretreatment* |
| | $\sum_i t_{ij} \leq 1$ | *use pretreatments at most once* |
| Outside subproblem | $\sum_k s_{ik} = 1$ | *all treatments get one substitute* |
| | $\sum_l p_{kl} \geq \sum_j s_{jk}$ | *every selected substitute gets a presub* |
| | $\sum_i s_{ik} \leq 1$ | *use substitutes at most once* |
| | $\sum_k p_{kl} \leq 1$ | *use presubs at most once* |

# IDENTIFICATION STRATEGY 2: QUARTET + X




Greedy optimization picks <hijl> for a cost of 11

Joint optimization picks <hikl> for a cost of 6!

Indeed, j might go unused w/o a nearer presub!

# IDENTIFICATION STRATEGY 2: QUARTET + X

$$\Upsilon_i - \Upsilon_h = \delta_i = f(X_i) + e_i$$ TREATED AREA MODEL

$$\Upsilon_k - \Upsilon_l = \delta_k = f(X_k) + e_k$$ UNTREATED AREA MODEL

$$\tau_1 = \delta_1 - \hat{\delta}_0 = f(X) + u$$ TREATED PO MODEL

$$\tau_0 = \hat{\delta}_1 - \delta_0 = f(X) + u$$ UNTREATED PO MODEL

$$p(t = 1) = f(X) + v$$ PROPENSITY MODEL

ATE: $n^{-1} \Sigma \, \tau_0 p + \tau_1 (1\text{-}p)$

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$$\Upsilon_k - \Upsilon_l = \delta_k = f(X_k) + e_k$$ UNTREATED AREA MODEL

*Predict the change in price for properties inside the DLA/outside of the DLA.*

# IDENTIFICATION STRATEGY 2: QUARTET + X

$$Y_i - Y_h = \delta_i = f(X_i) + e_i$$ TREATED AREA MODEL

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$$\tau_0 = \hat{\delta}_1 - \delta_0 = f(X) + u$$ UNTREATED PO MODEL

$\delta_1$: *Observed inside change*
$\hat{\delta}_0$: *Outside's prediction about an inside change*
$\delta_0$: *Observed outside change*
$\hat{\delta}_1$: *Inside's prediction about an outside change*

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*Predict the difference between observed change & counterfactual predicted change*

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$$p(t = 1) = f(X) + v$$ PROPENSITY MODEL

*How likely is a property with X treated?*

# IDENTIFICATION STRATEGY 2: QUARTET + X

$$\Upsilon_i - \Upsilon_h = \delta_i = f(X_i) + e_i$$ TREATED AREA MODEL

$$\Upsilon_k - \Upsilon_l = \delta_k = f(X_k) + e_k$$ UNTREATED AREA MODEL

$$\tau_1 = \delta_1 - \hat{\delta}_0 = f(X) + u$$ TREATED PO MODEL

$$\tau_0 = \hat{\delta}_1 - \delta_0 = f(X) + u$$ UNTREATED PO MODEL

$$p(t = 1) = f(X) + v$$ PROPENSITY MODEL

*Mean PO gap, weighted by treatment likelihood*

ATE: $n^{-1}\Sigma\ \tau_0 p + \tau_1 (1\text{-}p)$

## IDENTIFICATION STRATEGY 2: QUARTET + X

WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

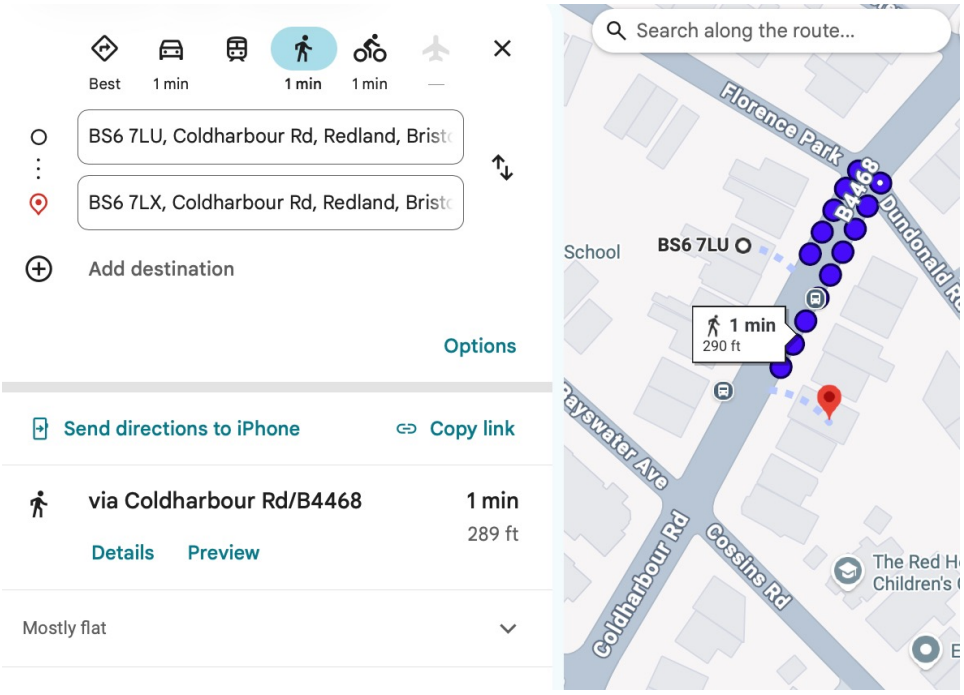
PRELIMINARY RESULTS

SLA OR NAY?

| kind | uid | Treated | In DLA? | Date | Postcode | Beds | Baths | Rent |
|---|---|---|---|---|---|---|---|---|
| **treated** | 7459f71781 | TRUE | TRUE | 2019-08-24 | BS67LX | 1 | 1 | 167 |
| **pretreat** | 777d5a2831 | FALSE | TRUE | 2018-07-28 | BS67LX | 1 | 1 | 162 |
| **substitute** | ea79ced515 | FALSE | FALSE | 2019-06-06 | BS67LU | 1 | 1 | 150 |
| **presubstitute** | cd861f2032 | FALSE | FALSE | 2018-06-29 | BS67LU | 1 | 1 | 144 |

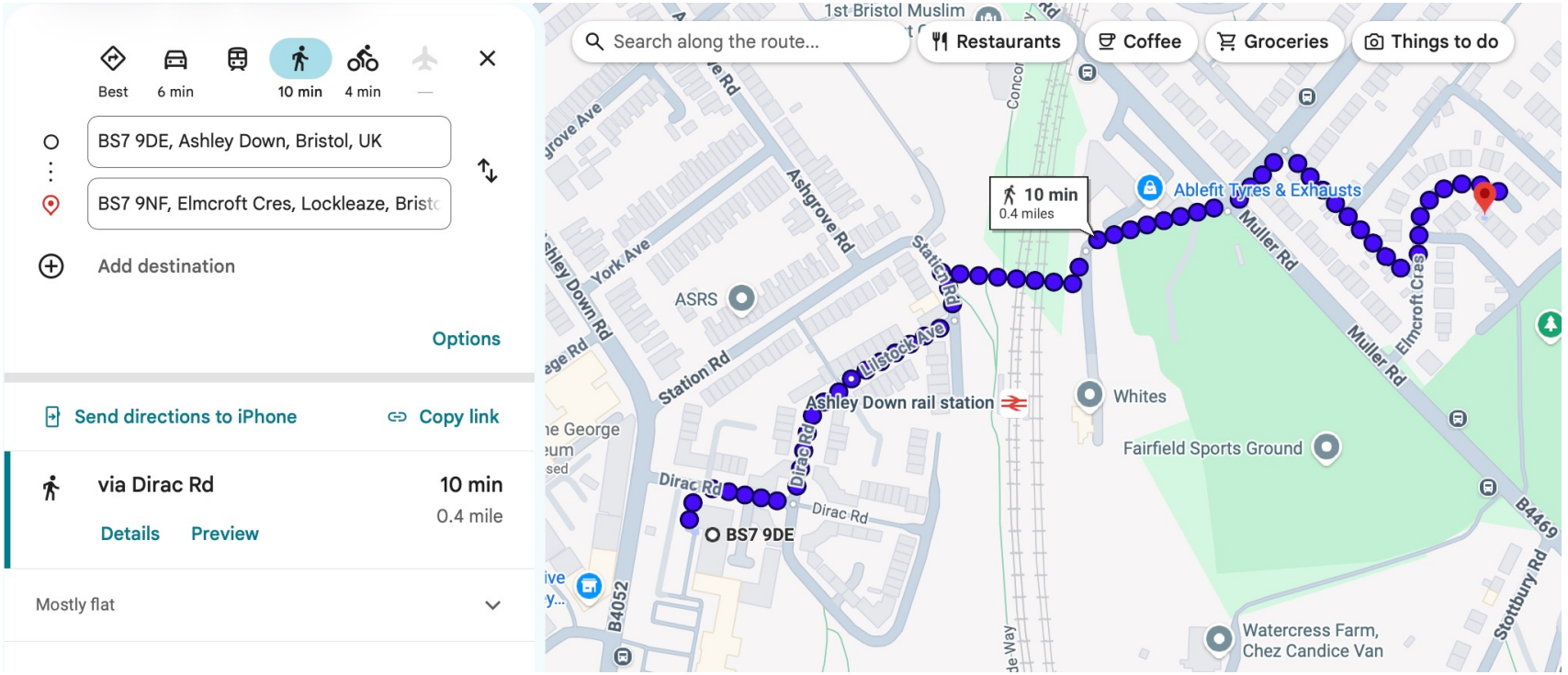
# MATCH EXAMPLE

| kind | uid | Treated | In DLA? | Date | Postcode | Beds | Baths | Rent |
|---|---|---|---|---|---|---|---|---|
| **treated** | 7459f71781 | TRUE | TRUE | 2019-08-24 | BS67LX | 1 | 1 | 167 |
| **pretreat** | 777d5a2831 | FALSE | TRUE | 2018-07-28 | BS67LX | 1 | 1 | 162 |
| **substitute** | ea79ced515 | FALSE | FALSE | 2019-06-06 | BS67LU | 1 | 1 | 150 |
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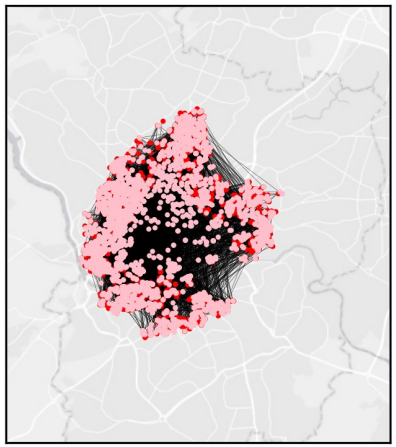

# MATCH EXAMPLE

| kind | uid | Treated | In DLA? | Date | Postcode | Beds | Baths | Rent |
|---|---|---|---|---|---|---|---|---|
| **treated** | 213afae5d7 | TRUE | TRUE | 2020-06-19 | BS79DE | 2 | 1 | 254 |
| **pretreat** | 2e11c99767 | FALSE | TRUE | 2019-04-25 | BS79DE | 2 | 1 | 231 |
| **substitute** | 6de3f08d32 | FALSE | FALSE | 2020-08-20 | BS79NF | 2 | 1 | 225 |
| **presubstitute** | db60128988 | FALSE | FALSE | 2019-07-12 | BS79NF | 2 | 1 | 219 |

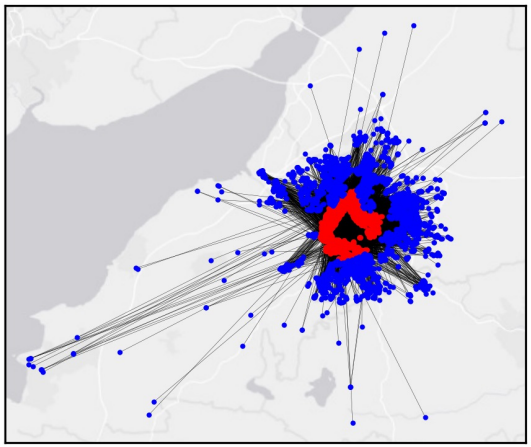



# MATCH EXAMPLE

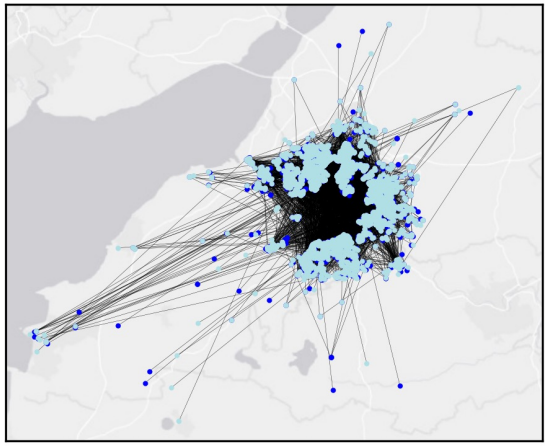
Treated to Predecessor

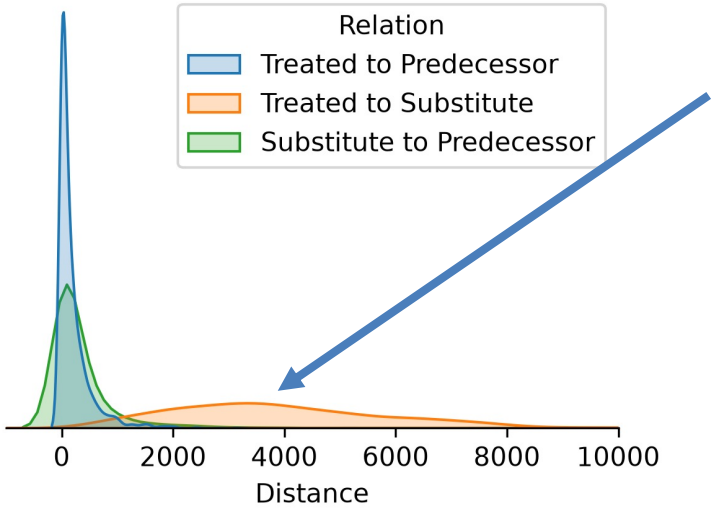


Treated to Substitute



Substitute to Predecessor



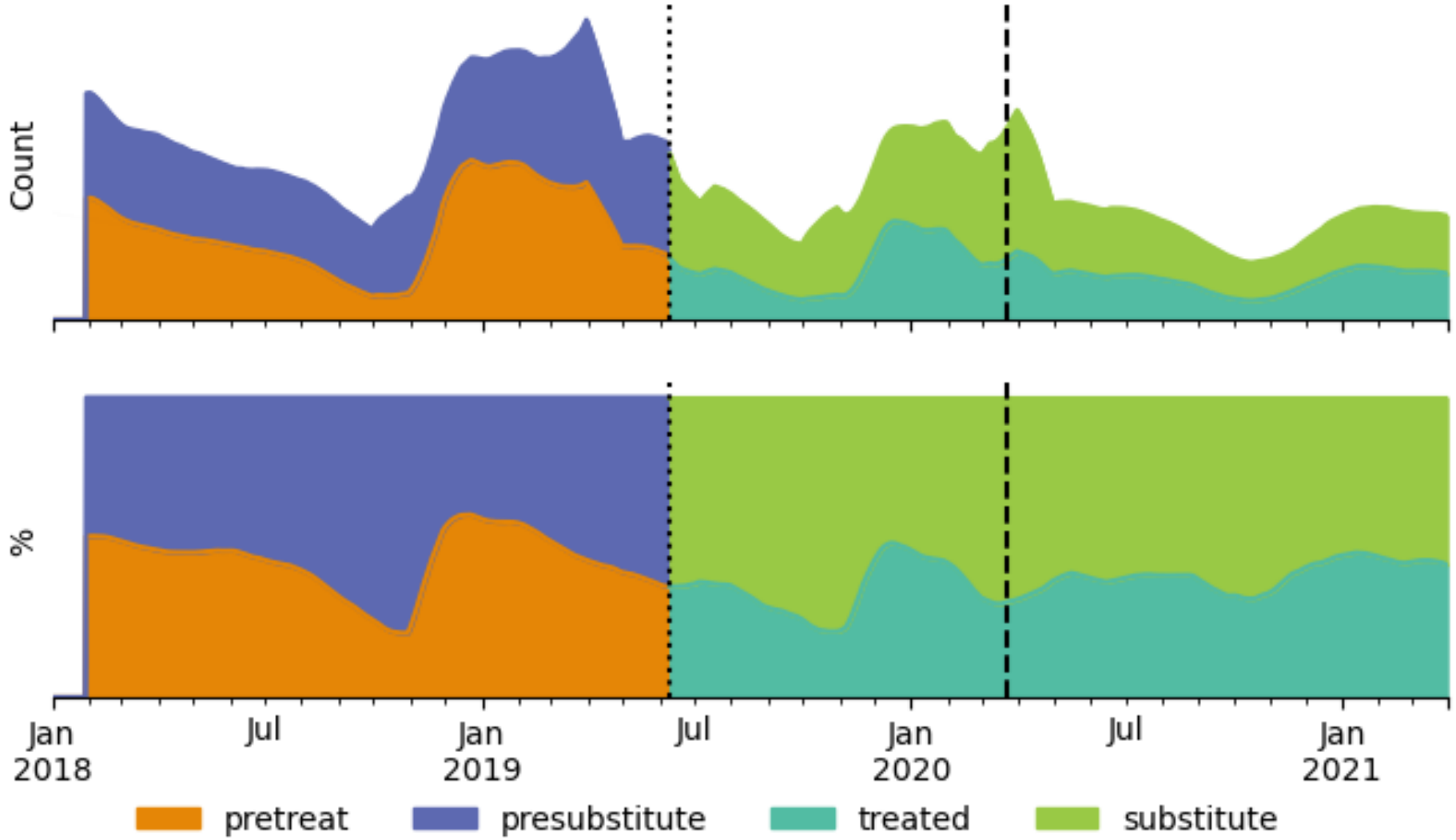



Proximity of the matching leaves something to be desired… but w/ ~5k matches, 729 are “local.”

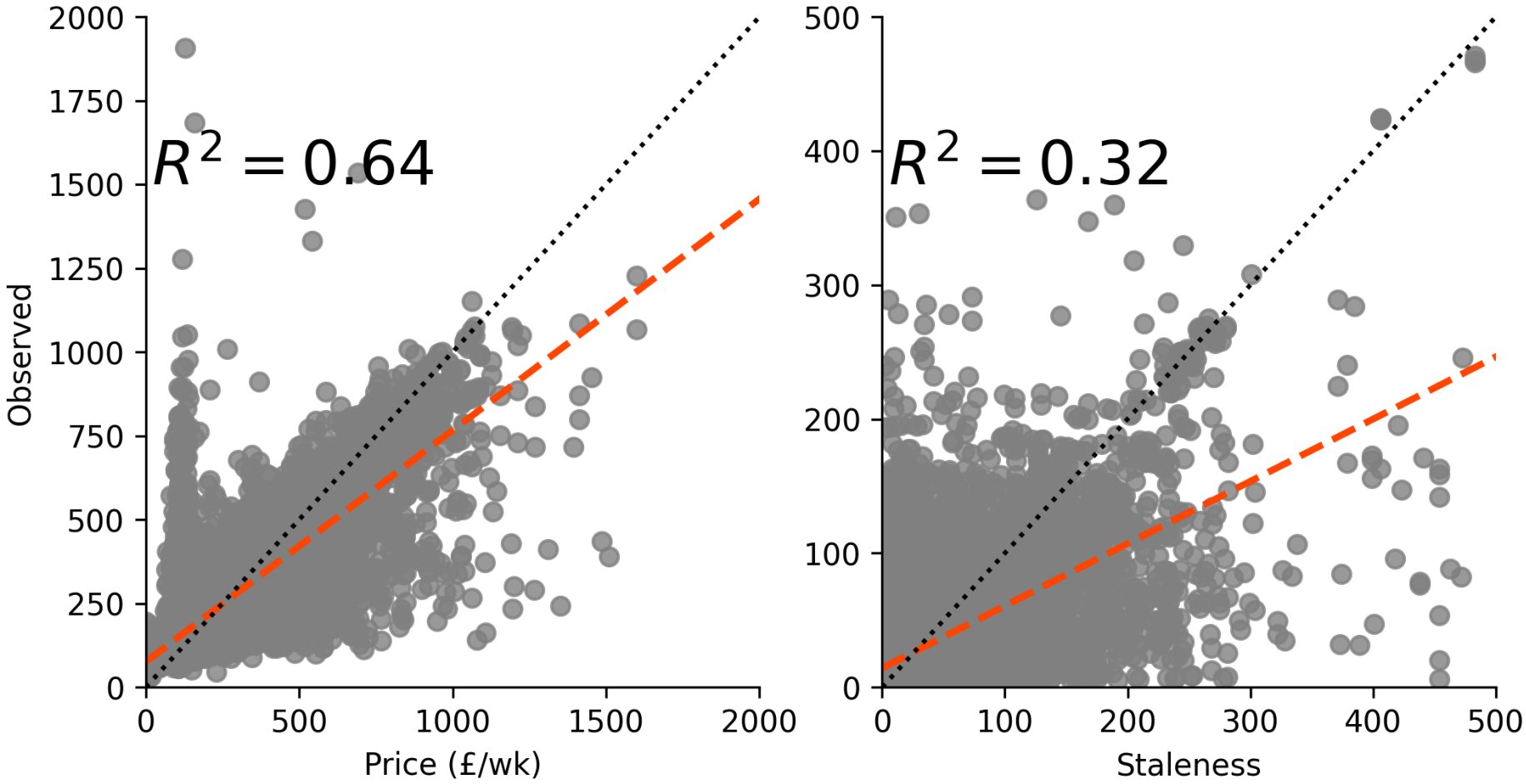
listings are 2018-2023
policy is 2019-2024

More data won’t solve the treated to substitute match

# MATCH QUALITY




# SUPPLY EFFECTS: TOTAL LISTINGS




# SPATIAL DIFFERENCE IN DISCONTINUITY

| | Overall | | Boundary | |
|---|---|---|---|---|
| **outcome** | duration | price | duration | price |
| **presubstitute** | 4.60 | -37.03 | -2.27 | -17.30 |
| **pretreat** | 4.04 | 26.15 | -0.38 | 20.49 |
| **substitute** | 2.51 | -42.05 | -6.33 | -7.98 |
| **treated** | 4.48 | 22.07 | -0.83 | 14.63 |

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| treated | 4.48 | 22.07 | -0.83 | 14.63 |

Predicted price from a listing's own-side model minus the across-boundary model

# SPATIAL DIFFERENCE IN DISCONTINUITY

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|---|---|---|---|---|
| outcome | duration | price | duration | price |
| **presubstitute** | 4.60 | -37.03 | -2.27 | -17.30 |
| **pretreat** | 4.04 | 26.15 | -0.38 | 20.49 |
| substitute | 2.51 | -42.05 | -6.33 | -7.98 |
| treated | 4.48 | 22.07 | -0.83 | 14.63 |

Models trained just inside the DLA boundary predict higher prices for the same houses…

# SPATIAL DIFFERENCE IN DISCONTINUITY

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| **outcome** | duration | price | duration | price |
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| **substitute** | 2.51 | -42.05 | -6.33 | -7.98 |
| **treated** | 4.48 | 22.07 | -0.83 | 14.63 |

Models trained just inside the DLA boundary predict higher prices for the same houses...
gap narrows post-treatment.

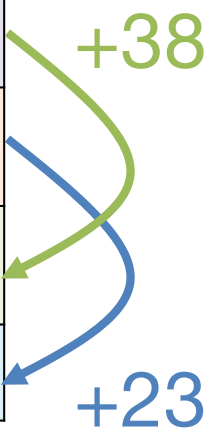
# SPATIAL DIFFERENCE IN DISCONTINUITY

| | Overall | | Boundary | |
|---|---|---|---|---|
| **outcome** | duration | price | duration | price |
| **presubstitute** | 4.60 | -37.03 | -2.27 | -17.30 |
| **pretreat** | 4.04 | 26.15 | -0.38 | 20.49 |
| **substitute** | 2.51 | -42.05 | -6.33 | -7.98 |
| **treated** | 4.48 | 22.07 | -0.83 | 14.63 |

| | Overall | | Boundary | |
|---|---|---|---|---|
| **outcome** | duration | price | duration | price |
| **ATE** | 2.52 | 0.94 | 3.62 | -15.17 |

# SPATIAL DIFFERENCE IN DISCONTINUITY

| | Overall | | Boundary | |
|---|---|---|---|---|
| **outcome** | duration | price | duration | price |
| **presubstitute** | 4.60 | -37.03 | -2.27 | -17.30 |
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| **substitute** | 2.51 | -42.05 | -6.33 | -7.98 |
| **treated** | 4.48 | 22.07 | -0.83 | 14.63 |




**CHANGE IN THE GAP**

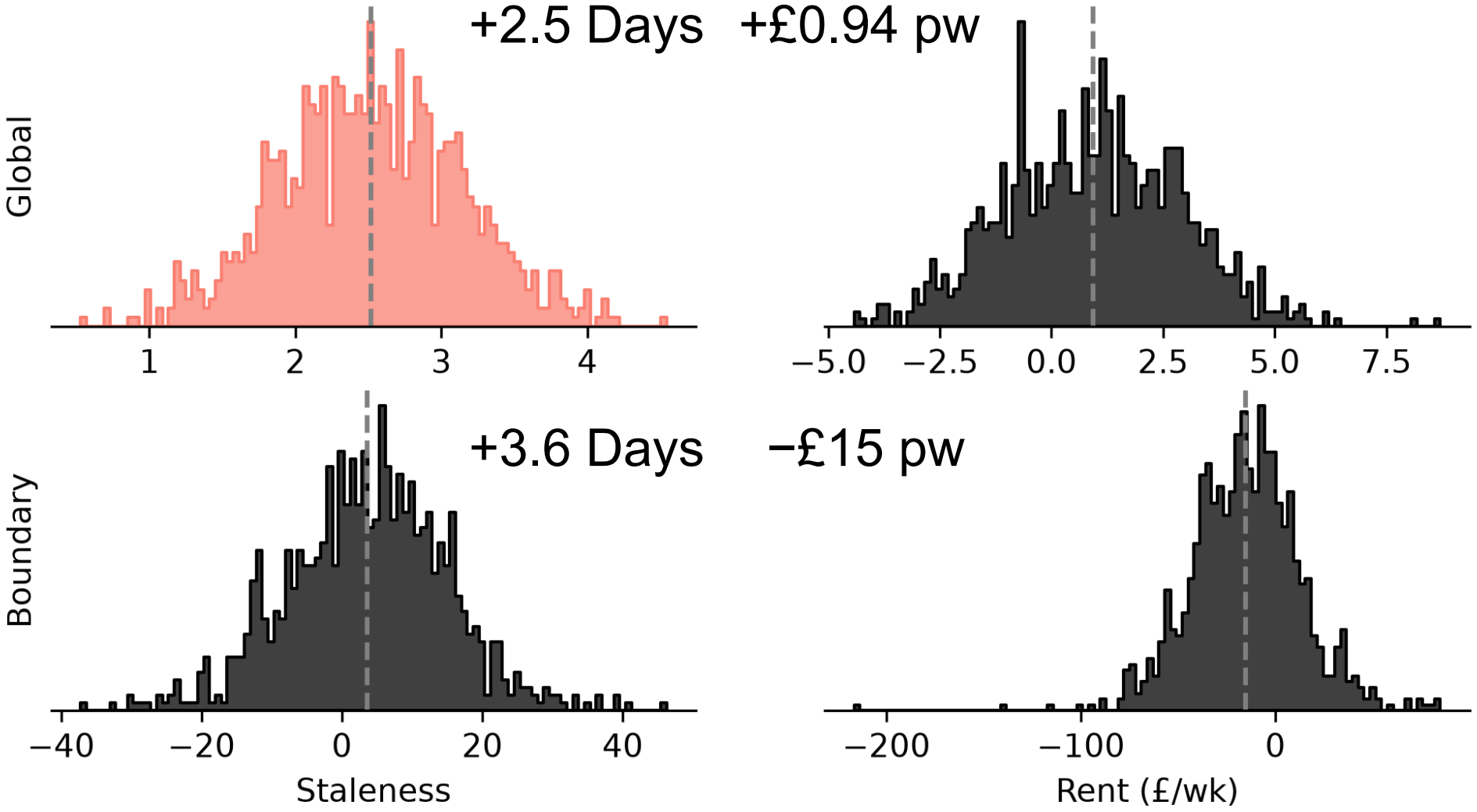
| | Overall | | Boundary | |
|---|---|---|---|---|
| outcome | duration | price | duration | price |
| ATE | 2.52 | 0.94 | 3.62 | -15.17 |

# SPATIAL DIFFERENCE IN DISCONTINUITY

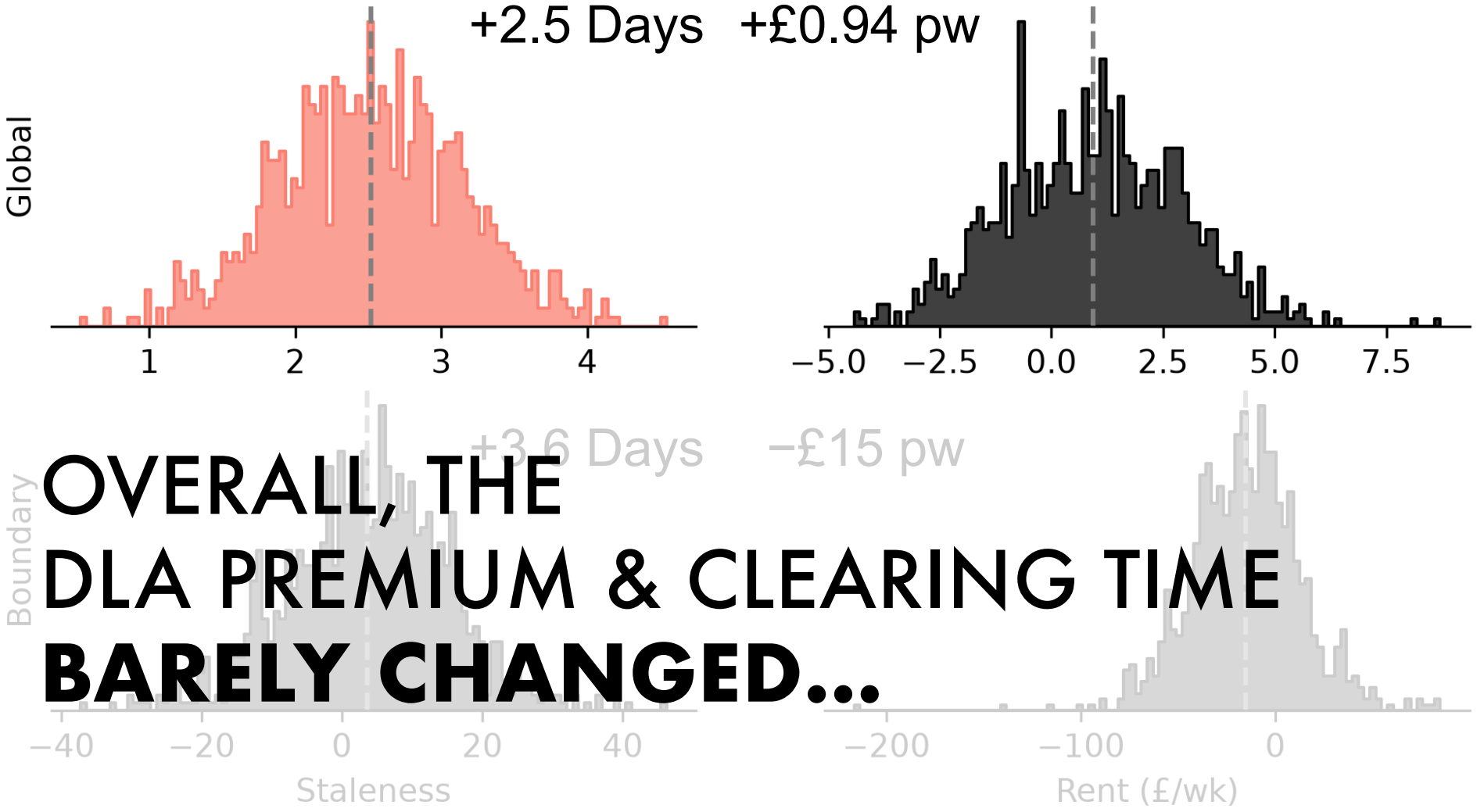
| | Overall | | Boundary | |
|---|---|---|---|---|
| **outcome** | duration | price | duration | price |
| **presubstitute** | 4.60 | -37.03 | -2.27 | -17.30 |
| **pretreat** | 4.04 | 26.15 | -0.38 | 20.49 |
| **substitute** | 2.51 | -42.05 | -6.33 | -7.98 |
| **treated** | 4.48 | 22.07 | -0.83 | 14.63 |

| | Overall | | Boundary | |
|---|---|---|---|---|
| **outcome** | duration | price | duration | price |
| **ATE** | 2.52 | 0.94 | 3.62 | -15.17 |

# SDIDc: SMALLER PREMIUM AT THE BOUNDARY



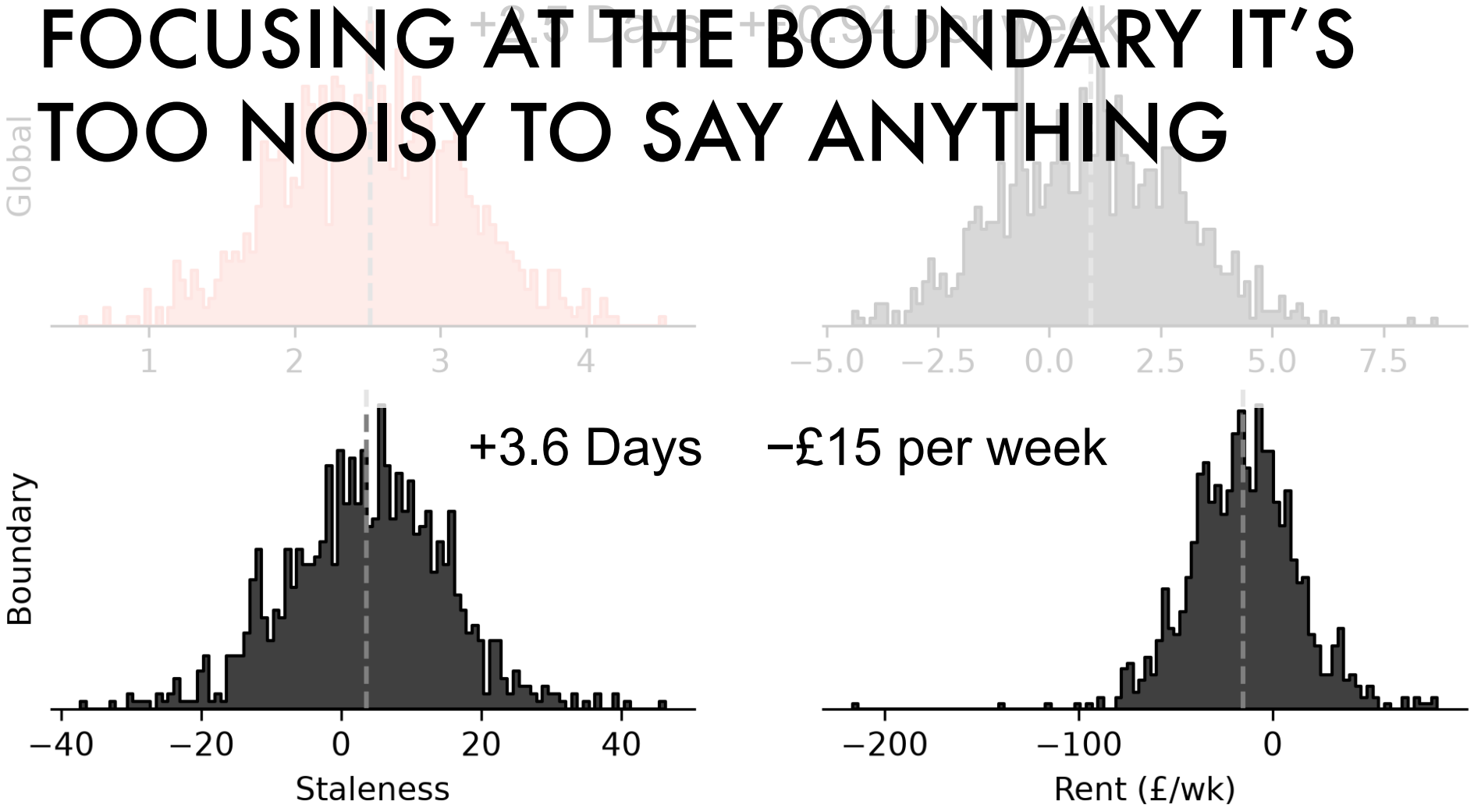

# SPATIAL DIFFERENCE IN DISCONTINUITY



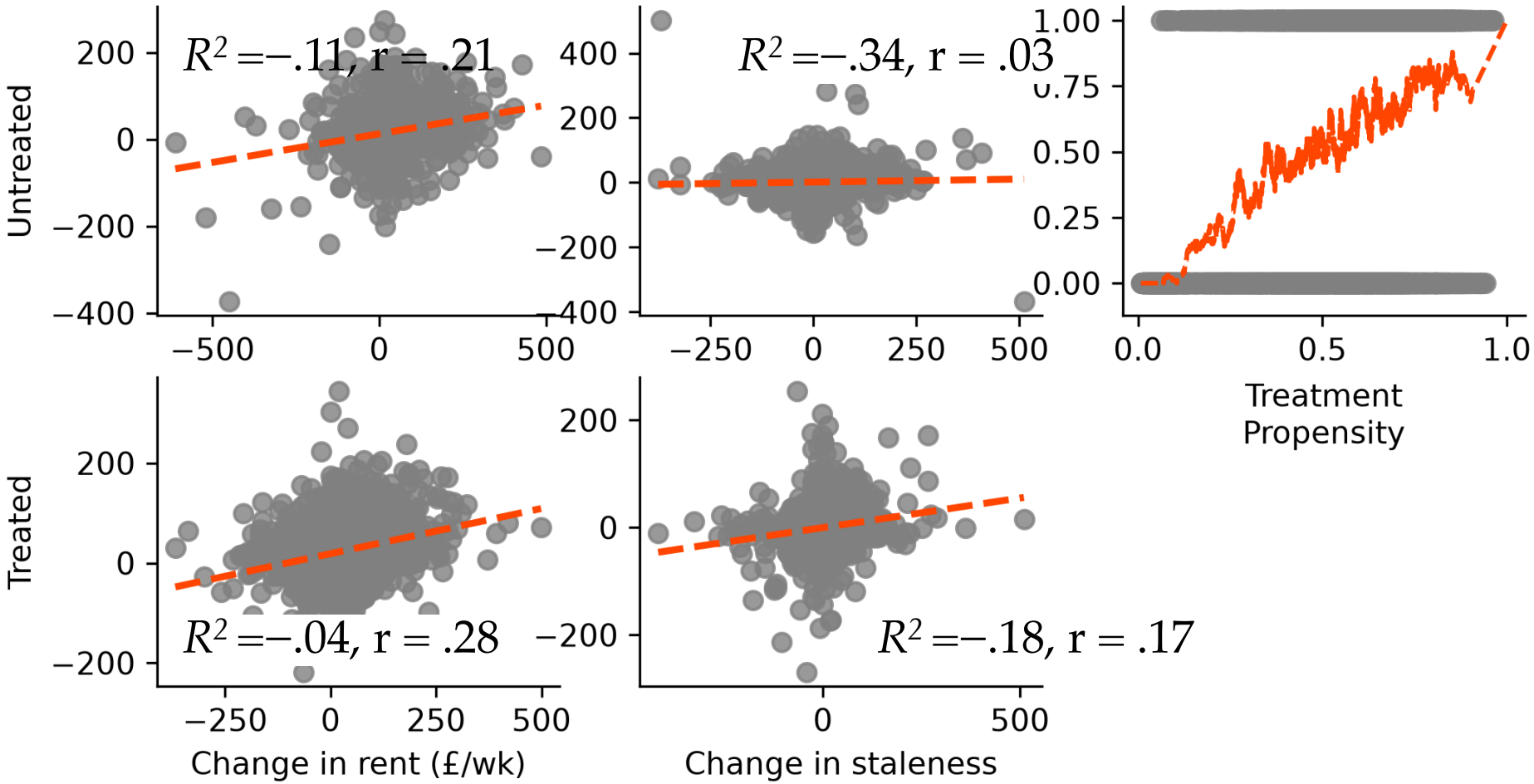

# OVERALL, THE DLA PREMIUM & CLEARING TIME **BARELY CHANGED...**

## SPATIAL DIFFERENCE IN DISCONTINUITY

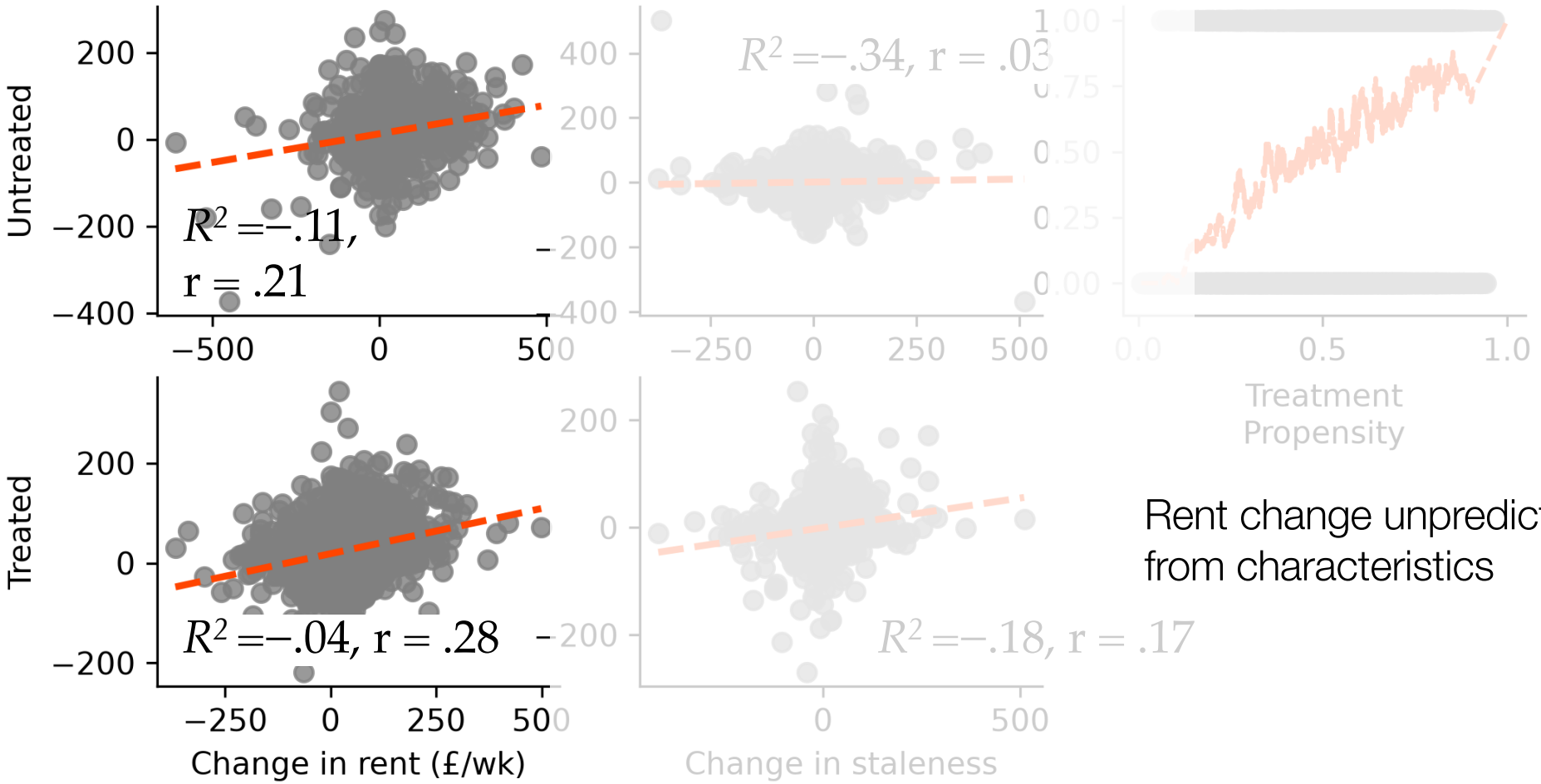
# FOCUSING AT THE BOUNDARY IT'S TOO NOISY TO SAY ANYTHING




## SPATIAL DIFFERENCE IN DISCONTINUITY

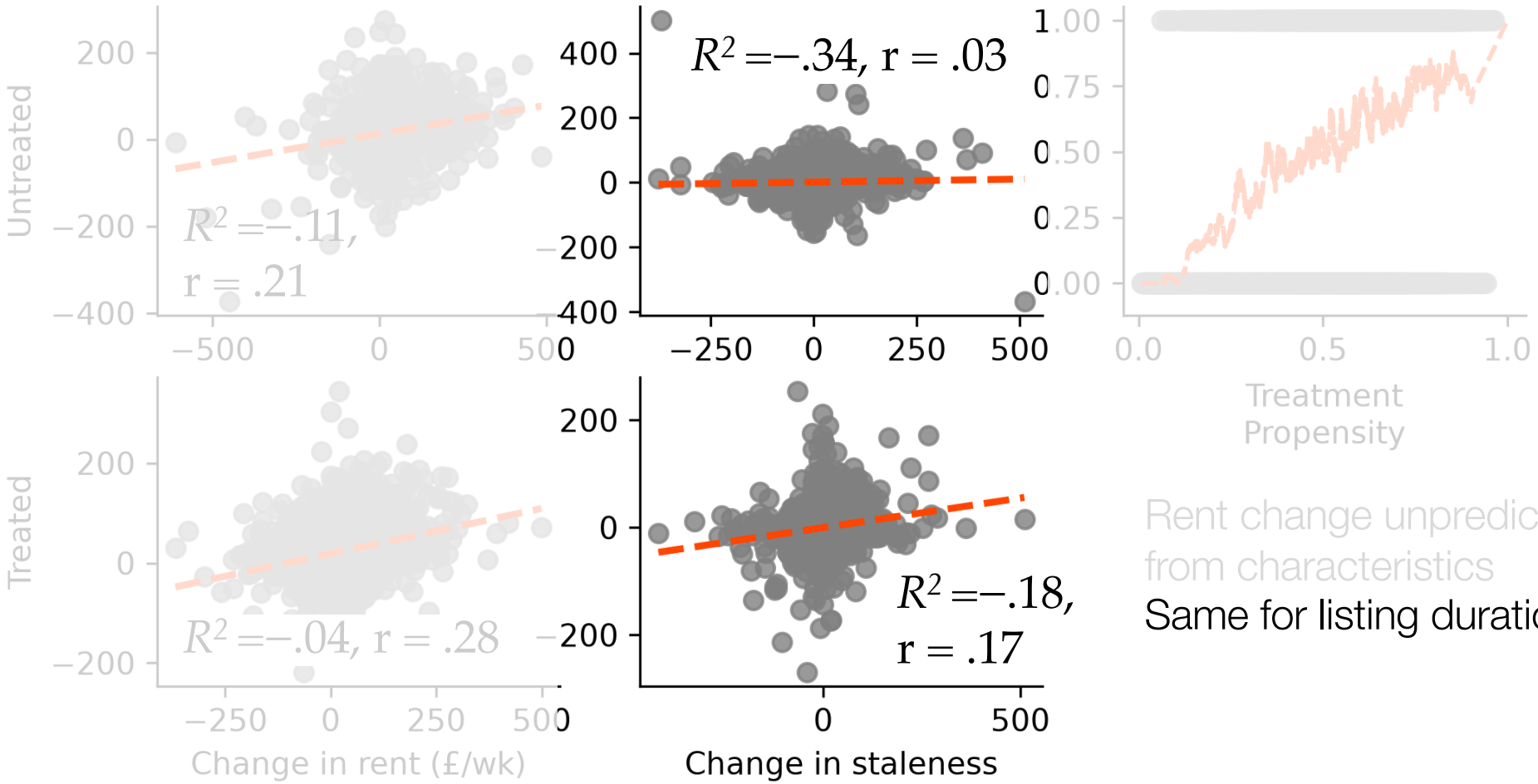

$R^2 = -.11$, r = .21
$R^2 = -.34$, r = .03
$R^2 = -.04$, r = .28
$R^2 = -.18$, r = .17
Untreated
Treated
Change in rent (£/wk)
Change in staleness
Treatment Propensity

# X-LEARNER




Rent change unpredictable from characteristics

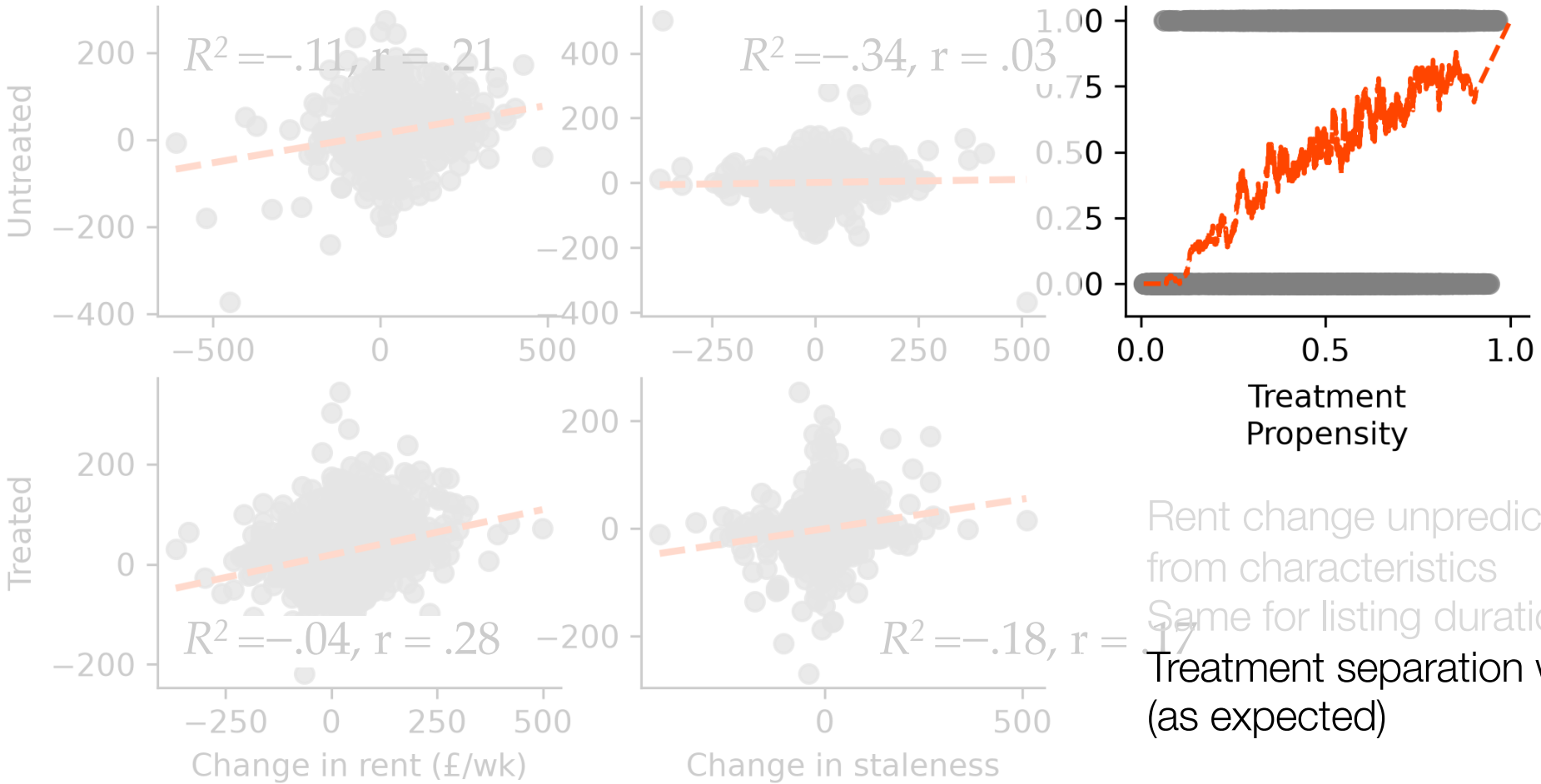
# X-LEARNER




Rent change unpredictable from characteristics

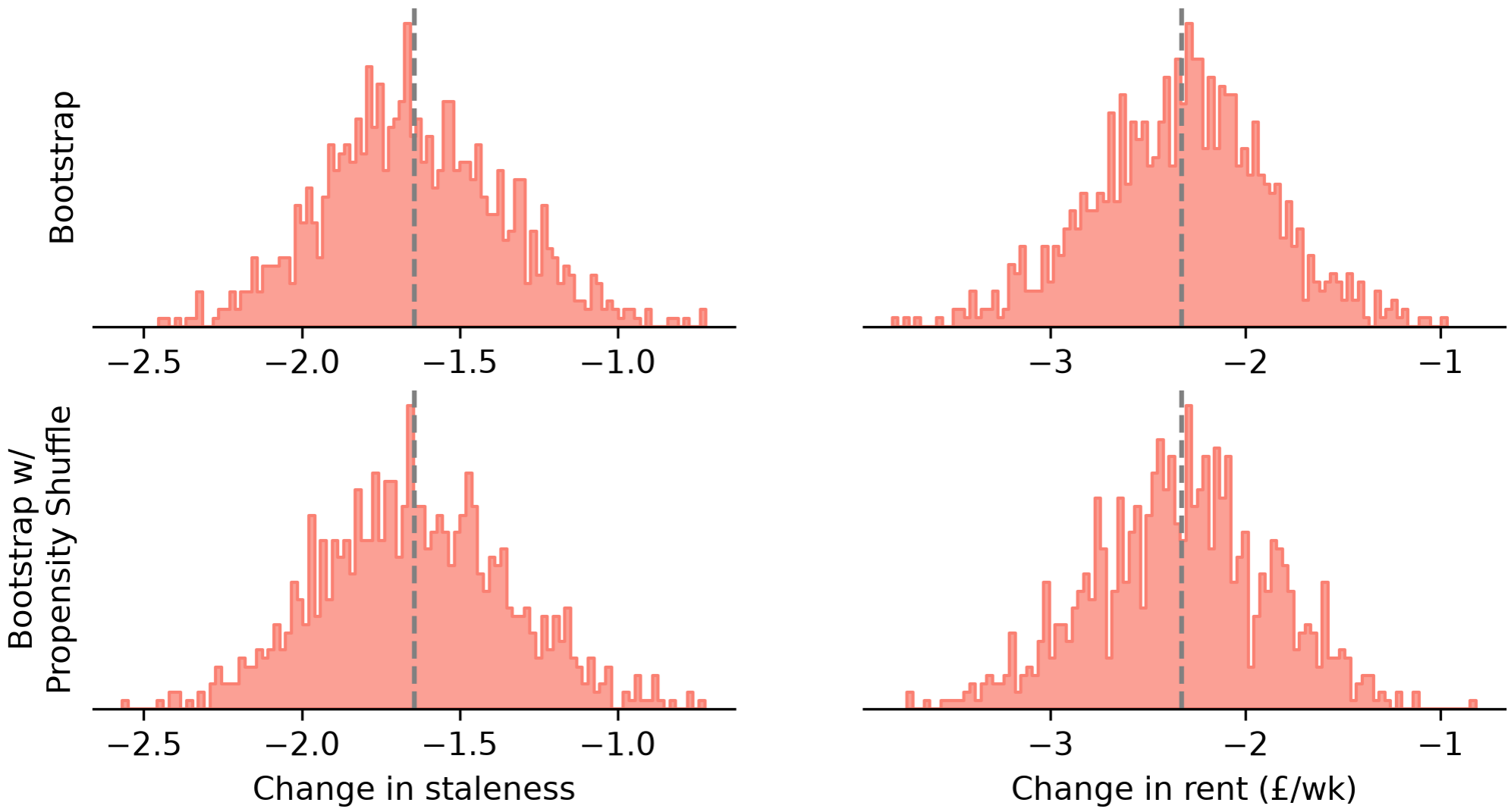
Same for listing duration…

# X-LEARNER

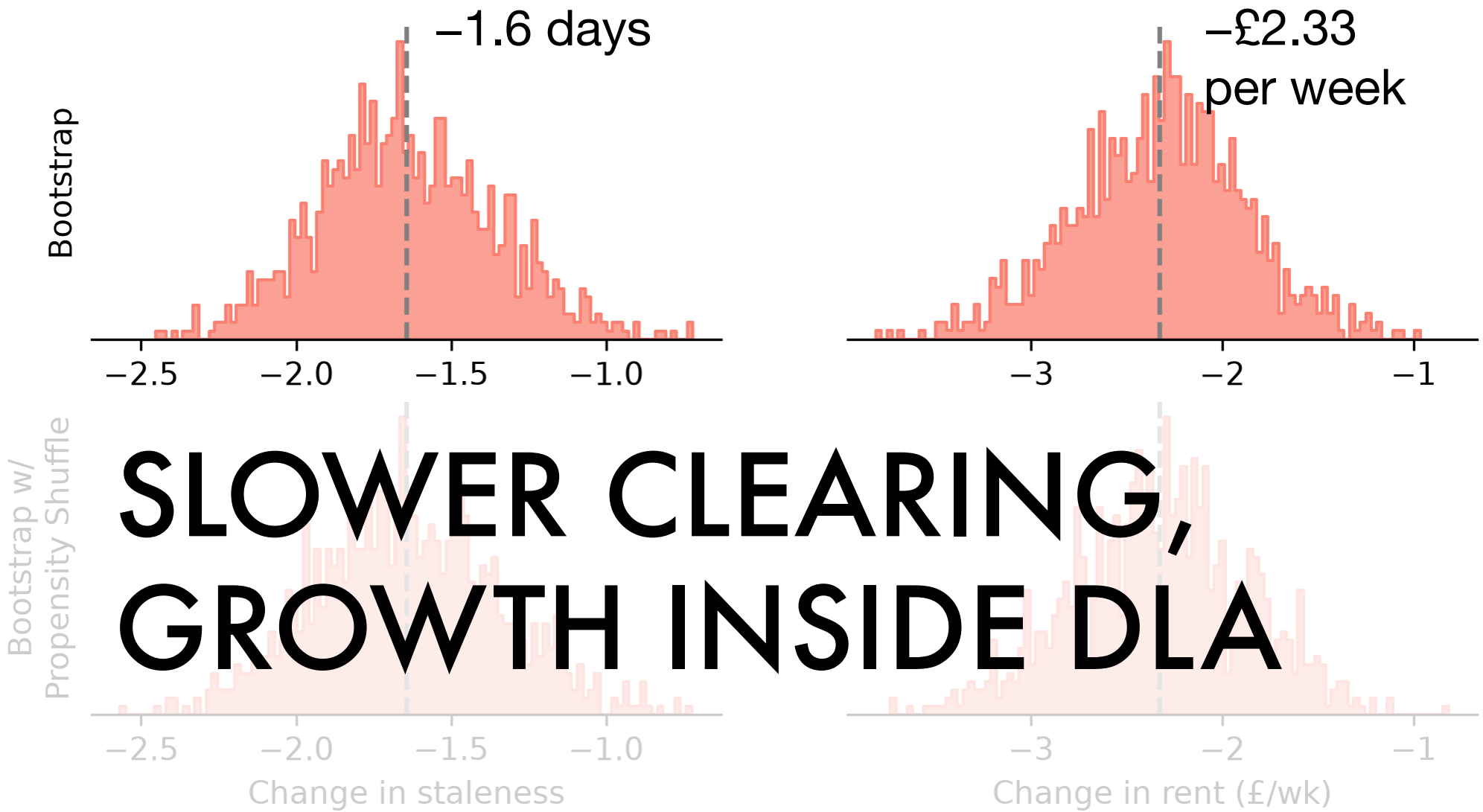



Rent change unpredictable from characteristics
Same for listing duration…
Treatment separation weak (as expected)

# X-LEARNER


Bootstrap
Bootstrap w/
Propensity Shuffle
−2.5
−2.0
−1.5
−1.0
−3
−2
−1
Change in staleness
Change in rent (£/wk)

# X-LEARNER




# SLOWER CLEARING, GROWTH INSIDE DLA

X-LEARNER

(**T** – **S**) - (**PT** – **PS**) = (**T** – **PT**) - (**S** – **PS**) = ATE

SDIDc X-Learner

True in theory, but empirically…

rent: SDIDc noisy negative, X-Learner negative

clearing: SDIDc noisy positive, X-Leaner negative

Why?

## SLA OR NAY?

(T – S) - (PT – PS) = (T – PT) - (S – PS) = ATE

SDIDc X-Learner

True in theory, but empirically…

rent: SDIDc noisy negative, X-Learner negative

clearing: SDIDc noisy positive, X-Leaner negative

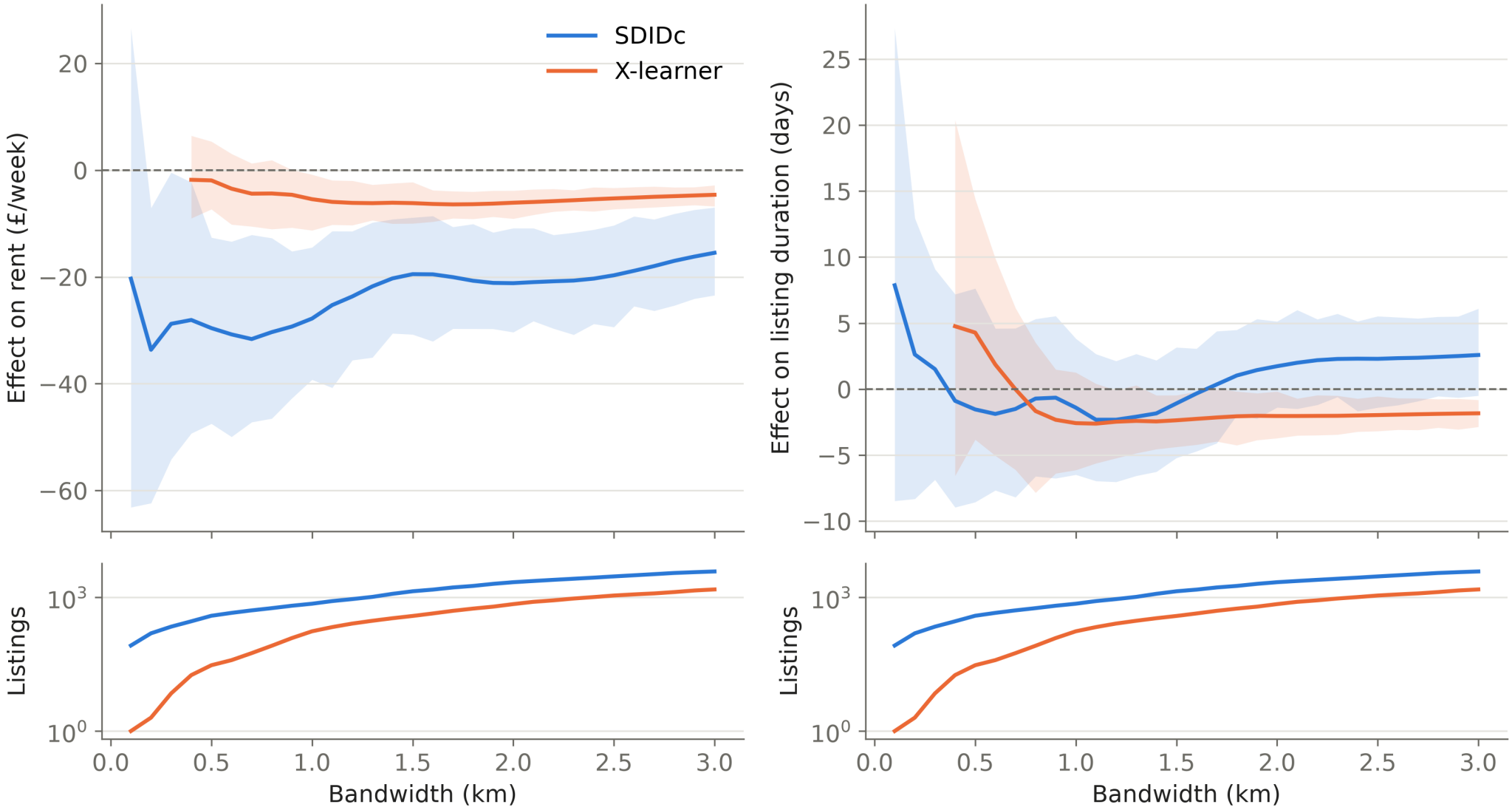
Why?

SDIDc: temporal difference of spatial differences

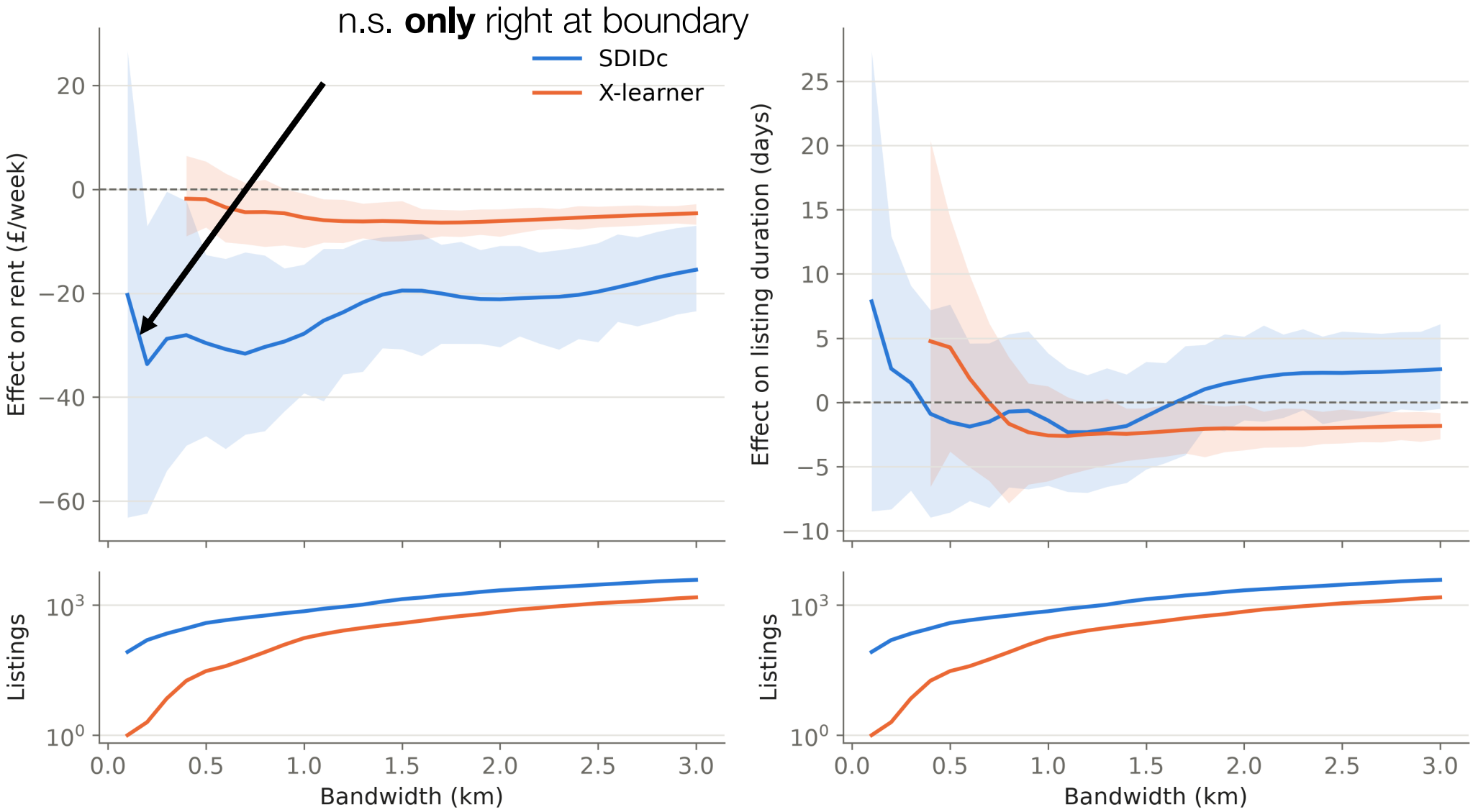
X-Learner: spatial difference of temporal differences

**How space gets differenced matters!**

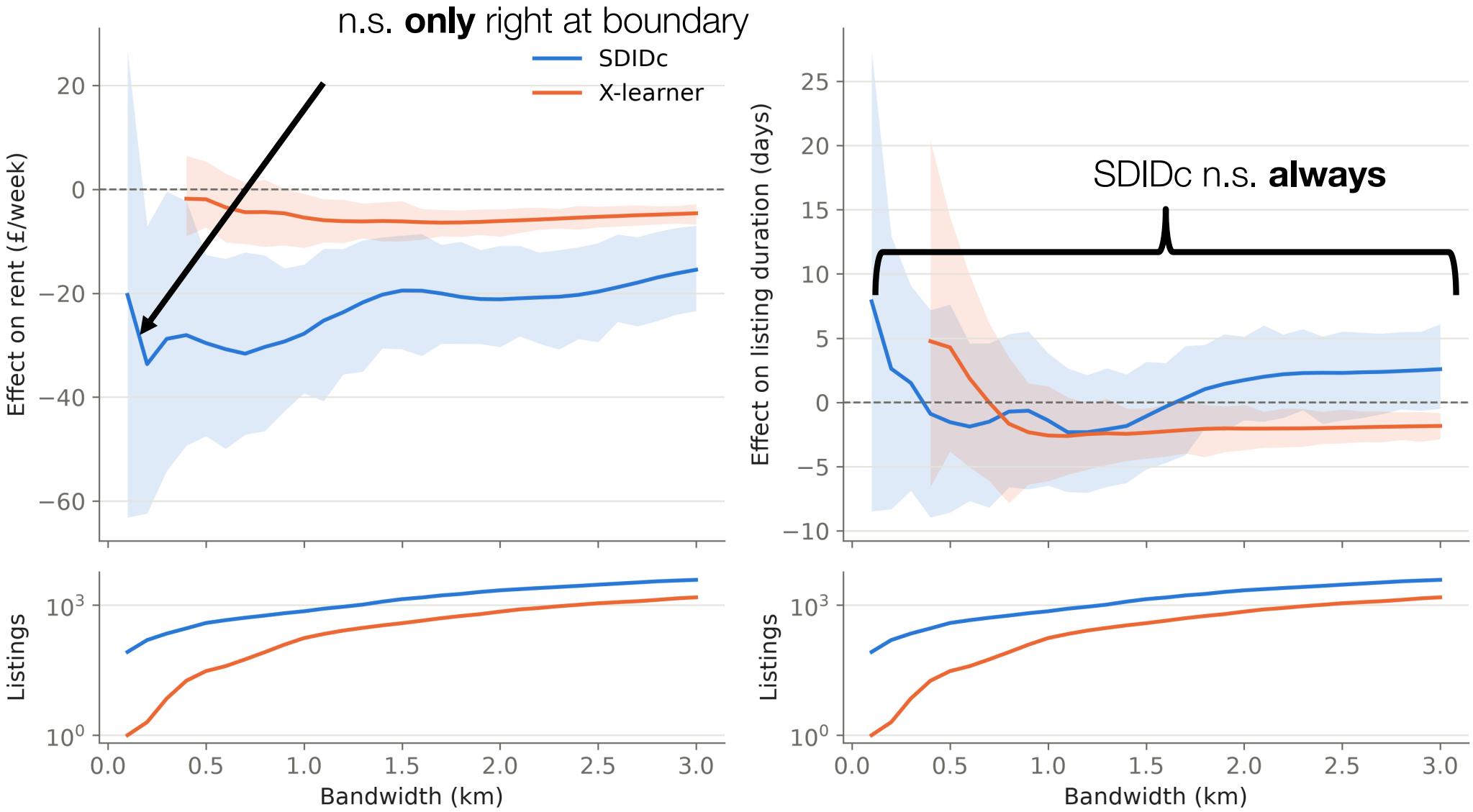
## SLA OR NAY?



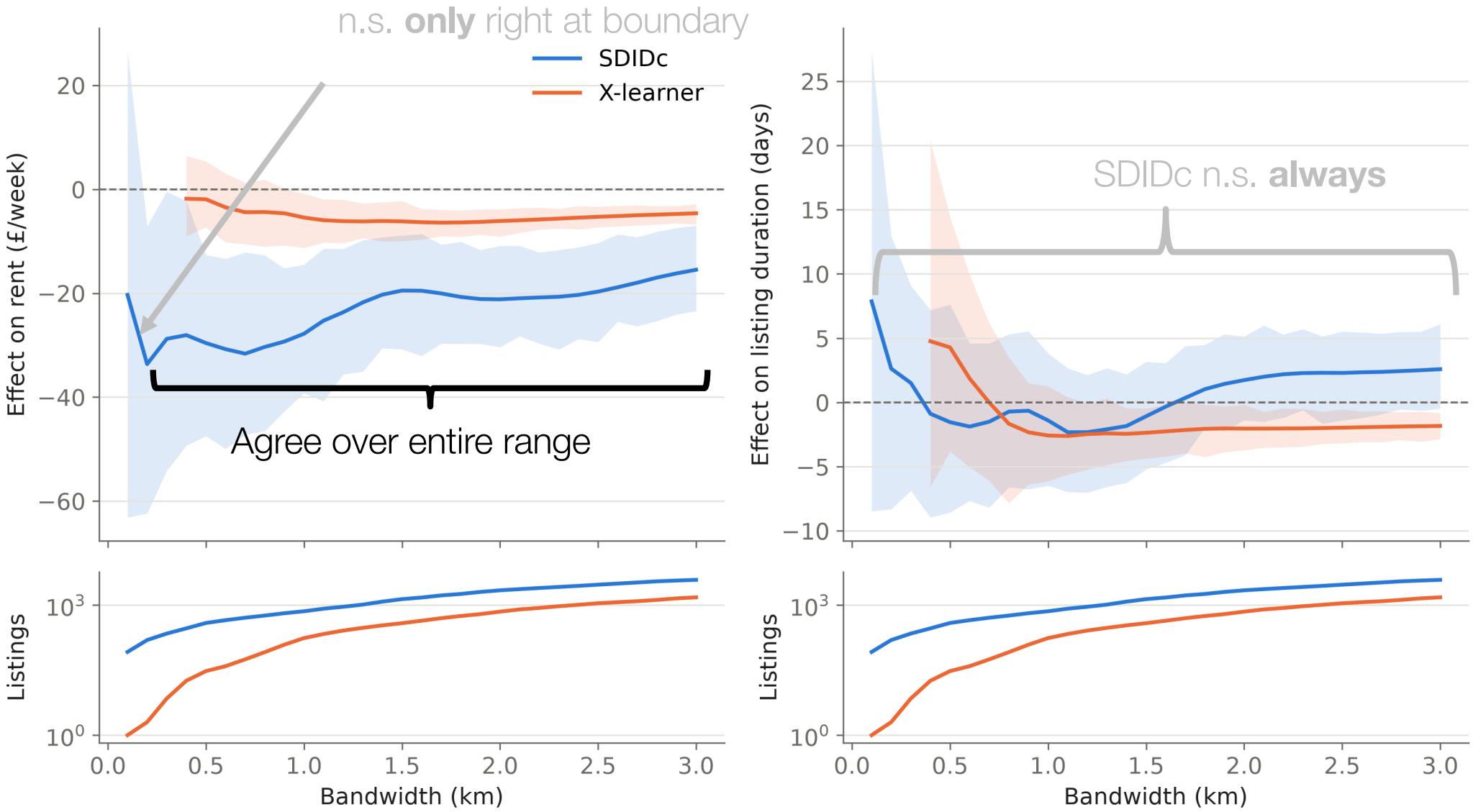

# SDIDc VS XLEARNER BY BANDWIDTH



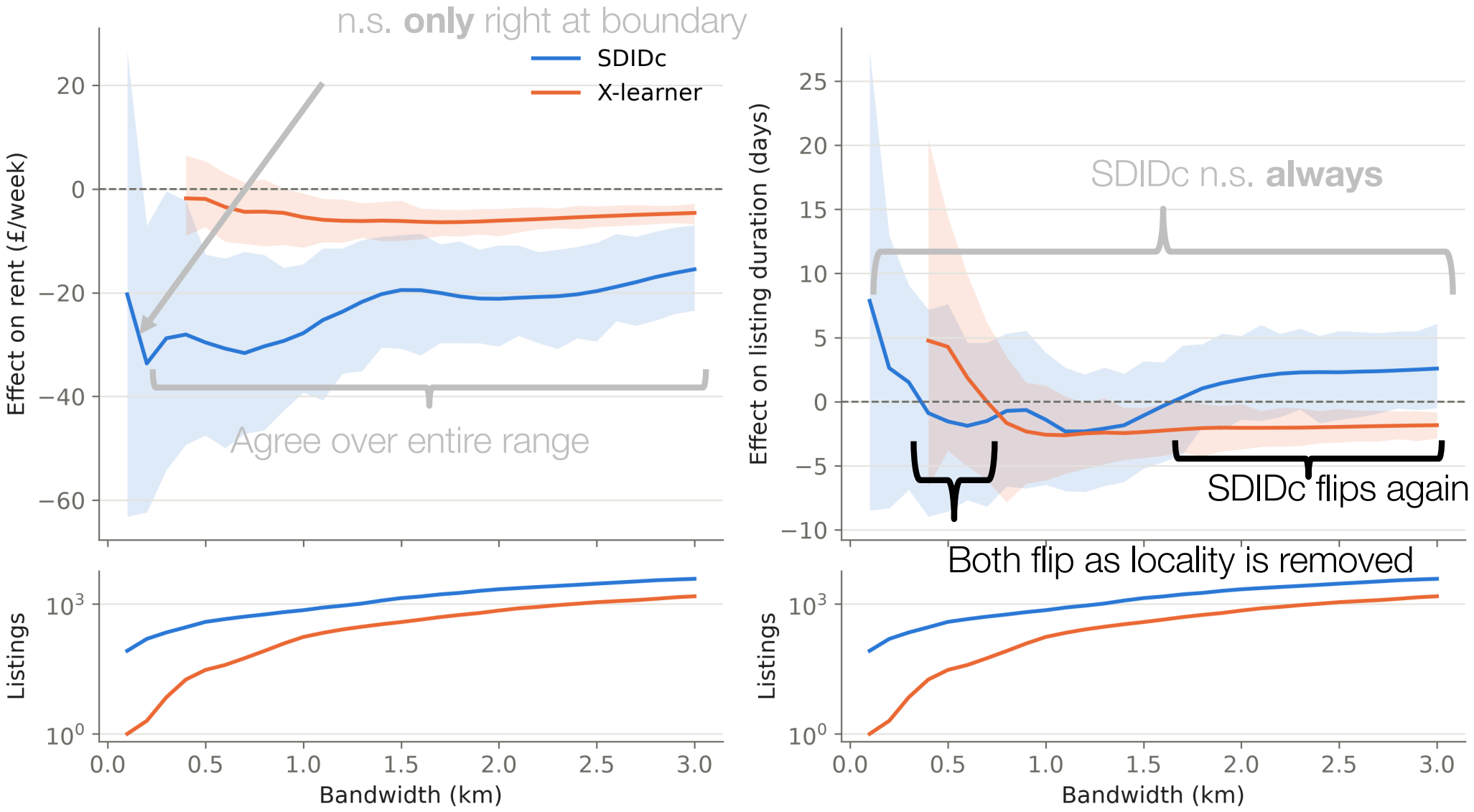

# SDIDc VS XLEARNER BY BANDWIDTH




# SDIDc VS XLEARNER BY BANDWIDTH




# SDIDc VS XLEARNER BY BANDWIDTH




# SDIDc VS XLEARNER BY BANDWIDTH

DLA rent premium narrowed between £20-£120, with no clear effect on clearing times, controlling for COVID-19, stock imbalance, path dependence.

Landlord concerns about fewer rentals & higher rents are not borne out in the short run.

Need access to property register to determine partial treatment, new rental market entrants

## SLA OR NAY?

# SLA OR NAY?

## CAUSAL IMPACTS OF BRISTOL DLAS

Levi John Wolf & Alex Marsh
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